

INSIDE PARAGRAPHS: typographic fundamentals

Cyrus Highsmith

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Written and illustrated by Cyrus Highsmith

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This book is dedicated to Sam Berlow.

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WE LIVE in a typographic wonderland. Typographers have more typefaces to choose from than ever before, and computers make setting type easier than ever. The truth is, if you're using a good typesetting application and you just leave the settings on default, you can set type that's more or less adequate.

However, good typographers are experts who have something to add to the reading experience. They finely tune the typography for specific documents to make it more than just adequate. In their choice of typefaces and decisions about things like point size and spacing, typographers clarify the voice of the author and make the reading experience more enjoyable.

This isn't meant to discourage beginners from attempting typography. But it is important to acknowledge that the art of typography is often subtle, and without training it can be difficult to see what's going on. The goal of this book is to help students train their eyes to see text as typographers do.

INTRODUCTION

The focus of this text is on the Latin alphabet (which is the alphabet used for English) and Latin typography. Of course, there are many other writing systems in use in our world. There are other alphabetic systems like Greek, Hebrew and Arabic. There are syllabic systems like Hangul, used for the Korean language. There are logographic systems like Chinese. The rules for Latin typography might not apply to these other writing systems.

This doesn't mean that our typographic insights are limited to English. Variants of the Latin alphabet are used all around the world to write hundreds of different languages. These include Roman languages like French, Spanish and Italian; Germanic languages like English, Dutch and German; Celtic languages like Irish and Welsh; Slavic languages like Czech and Polish; and many others. Some of these languages use characters that are not used in English, like the Danish Ø or the German ß. Or characters might be dropped, like the K in Italian. The Latin alphabet is a global phenomenon.

Even for languages within the same writing system, different rules may apply. In addition to the basic rules of Latin typography that they share, there are specific rules tuned to the peculiarities of a language's grammar and appearance. The focus here is mostly limited to the common basics.

This book is about what goes on inside a paragraph of printed text, and there are three different ways to read it: you can just look at the illustrations (which is how I would start), you can just read the text, or you can do both.

It begins with general explanations of how type works and how we read. Then it steps through the different kinds of space in a paragraph. Finally, it puts everything together with a discussion about paragraph settings.

Typography started with paragraphs of printed text. Since then, it has evolved in all sorts of directions, sometimes leaving printing behind entirely. But the printed paragraph is still a good starting point. Understanding what goes on inside it is a solid foundation on which to add additional knowledge.

And that's what this book is meant to be — a foundation. It's the book I wanted when I was a student in my first typography course.

how type works

HOW TYPE WORKS

COMPUTERS HAVE MADE SETTING TYPE fast and easy compared with hand-setting metal type (also known as foundry type). But many of the basic concepts of typesetting are still the same. A good way to learn how to use typesetting software and understand how type works is to go back to the very beginning, to the invention of movable type.

But this isn't exactly a historical overview. Instead, we'll imagine what the thought processes that lead to the invention of movable type might have been. The creative leap from hand-lettered paragraphs to paragraphs of printed text will help illustrate how type works.

It can be difficult to understand how something works just by reading about it, but don't let that slow you down. The more you practice setting type, the better you'll understand many of the concepts in this section. As you practice composing different docu-

ments and get to know more about typography by working with type, read this part again a few times. The goal is to learn what's going on behind the scenes when you set a paragraph. The better you understand these concepts, the better you'll be able to control what happens in your typography. And the more type you set, the better you'll understand these concepts.

Setting type can be thought of as a collaboration between the typographer and the typeface. A good typographer understands the typeface's role in this process, in part by understanding how movable type works. But it isn't necessary to learn all the complicated technical details of typesetting software. The important concepts can be learned by imagining how movable type was invented.

eneisquaecapitacu
em caelaturis su s
labe buntage ente as
12
imiliteret inlateral

HOW TYPE WORKS

Gutenberg's Idea

Johann Gutenberg of Mainz, Germany, is usually given credit for inventing movable type around the year 1450. This invention is the basis for the type we use today. Let's imagine what Gutenberg might have been thinking.

Gutenberg didn't invent books, and his invention didn't immediately change how a book looked to the reader. He invented a new path for creating a paragraph of text. Instead of writing out each line by hand, drawing each word and each letter, Gutenberg came up with a way of simulating a handwritten paragraph.

A piece of foundry type illustrates the gist of Gutenberg's idea: to put each letter into its own box. Putting odd-shaped things into standard-sized boxes makes them easier to handle. Think about carrying four pairs of shoes, compared with carrying four pairs of shoes in shoeboxes.

This is basically what Gutenberg did with movable type. Dealing with a paragraph of text is more complicated than carrying a pile of shoes, but the concept is similar. Gutenberg put the letters into boxes so they'd be easier to pick up and arrange.

COUNTER SPACE

LETTER SPACE

LINE SPACE

aeneisqua
cem caelato
habebunta
cimitorot

HOW TYPE WORKS

Gutenberg looked at the basic elements that make up a paragraph. Then he thought about how these elements interact. Finally, he created a way of building a paragraph using these interactions in a new way — a typographic way.

A paragraph has two basic visual elements: the black parts and the white parts. The black is the letters, the white is background. But instead of thinking of them as separate layers, like ink and paper, imagine the black parts and white parts together, like two parts of a puzzle.

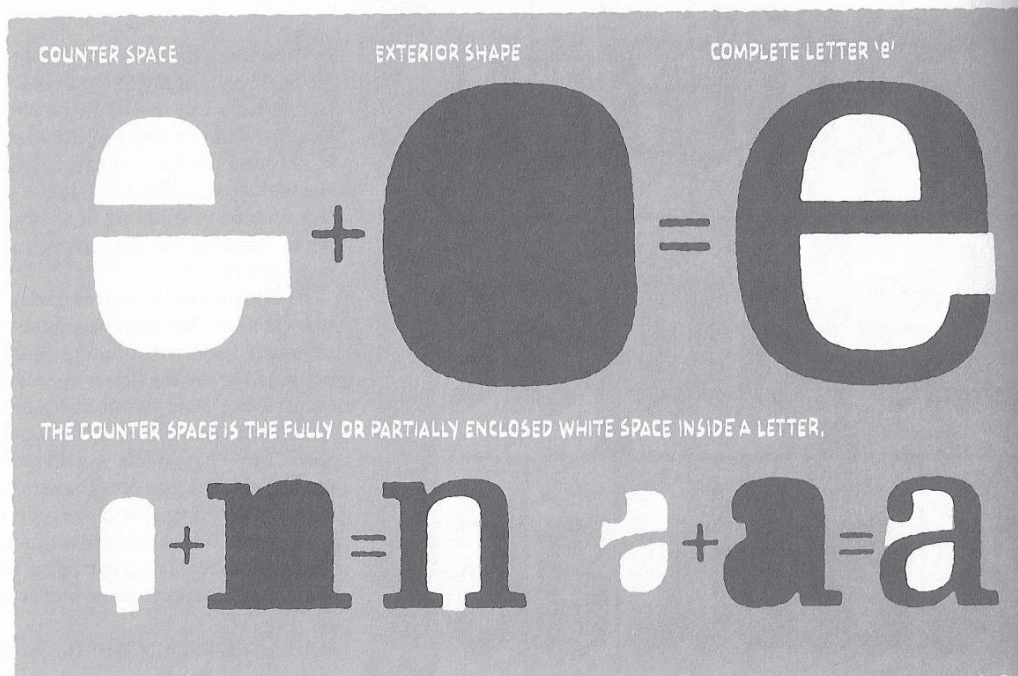
The black letters are arranged to form words, which are arranged to form lines. The white elements — the other puzzle pieces — are the white space within a letter (or counter space), the white space between the letters (letter space), the white space between words (word space), and the white space between lines (line space).

Gutenberg considered the counter space, letter space, and line space. Every paragraph, whether written or printed, has these white spaces in it. But they don't have to be thought of in isolation. Gutenberg's idea was to attach a certain amount of each kind of space to each letter. With this innovation he created a new kind of space: the glyph space. The glyph space forms the box around each letter that makes it easy to move and rearrange.

GLYPH SPACE: THE BOX THAT CONTAINS EACH CHARACTER,



LETTERS ARE MADE OF BOTH
BLACK AND WHITE SHAPES.



Et columnas vi intic
un basibus tot demi
aeneisquaecapitacu
cem caelaturis su se
habebuntageenteas
similiteretinalateral
ilonisperlongumeri
tentoriacentumcubi

HOW TYPE WORKS

Counter Space

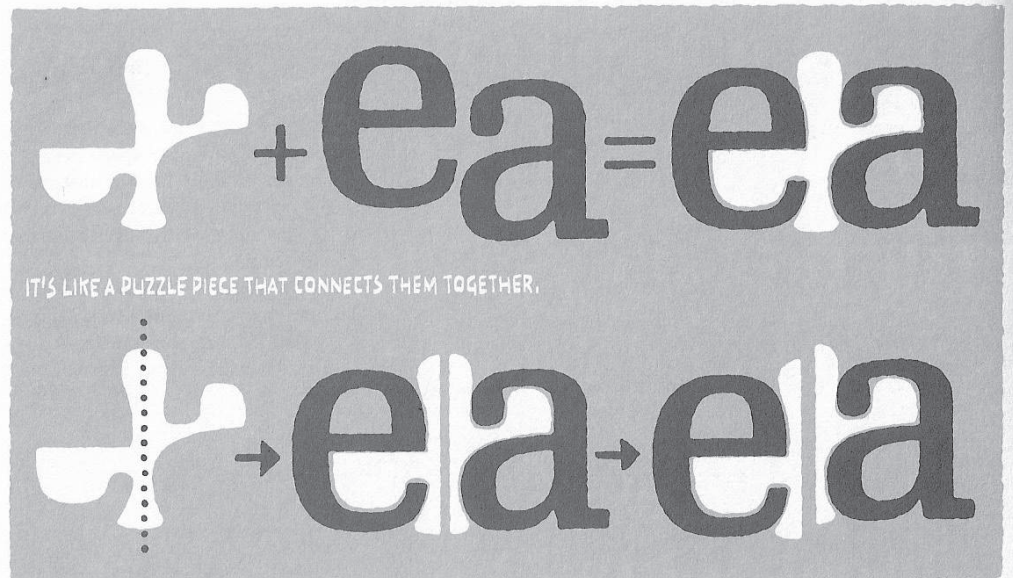
The counter space is the fully or partly enclosed white space inside a letter.

Counter space is similar to the empty space inside a bowl or container. The empty space isn't a physical thing like the material of the bowl itself. But the sides of the bowl cause a certain amount of space to be created inside that bowl. When you move the bowl, the space moves with it. Without that space, the bowl just wouldn't be the same.

The counter space of a letter belongs to that letter just like the empty space inside a container belongs to that container. When you move a letter, the counter space moves with it. Without the counter space, the letter just wouldn't be the same.

THE STYLE OF CALLIGRAPHY THAT GUTENBERG COPIED FOR THE FIRST
MOVABLE TYPE WAS CALLED TEXTURA.

LETTER SPACE IS THE WHITE SPACE BETWEEN LETTERS.



IT'S LIKE A PUZZLE PIECE THAT CONNECTS THEM TOGETHER.

THE LETTER SPACE CAN BE DIVIDED SO A CERTAIN AMOUNT 'BELONGS' TO EACH LETTER AND BECOMES PART OF THE GLYPH SPACE.

Et columnas vii intic
un basibus tot demi
aeneis quaecapitacu
cem caelaturis su se
habebunt age ente as
similiter et in laterial
ilonis per longumeri
tentoriacentum cubi

Letter Space

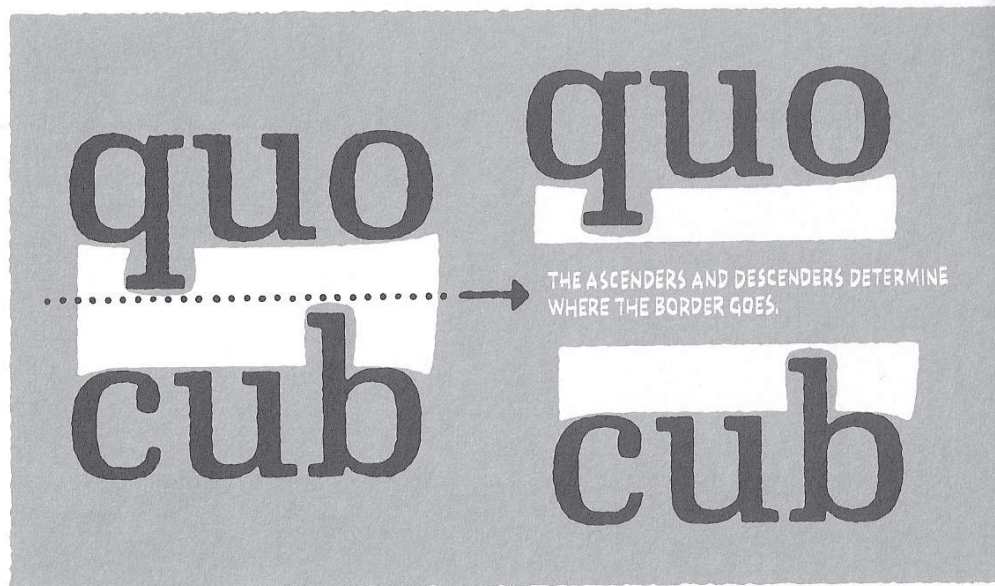
The letter space is the white space flanking letters on the left and right sides. In contrast to the counter space, which is interior space, letter space is exterior space.

In a letter like the c, the interior counter space is only partially enclosed and it flows into the exterior letter space. In a letter like the o, the counter space is fully enclosed and is distinct from the letter space.

A straight border can be imposed between each letter, splitting the white space so that a bit belongs to the letter on the left and a bit belongs to the letter on the right. These simple borders form the left and right sides of the glyph space.

TEXTURA'S STEADY RHYTHM MAKES IT EASY TO IMAGINE HOW IT COULD BE MADE INTO A MODULAR SYSTEM.

LINE SPACE IS THE WHITE SPACE BETWEEN THE LINES. IT'S THE PUZZLE PIECE THAT CONNECTS THEM TOGETHER.



A STRAIGHT BORDER CAN SPLIT THE LINE SPACE SO A CERTAIN AMOUNT BELONGS TO THE LINE ABOVE AND A CERTAIN AMOUNT BELONGS TO THE LINE BELOW.

Line Space

The line space is the white space above and below letters, formed between lines of text.

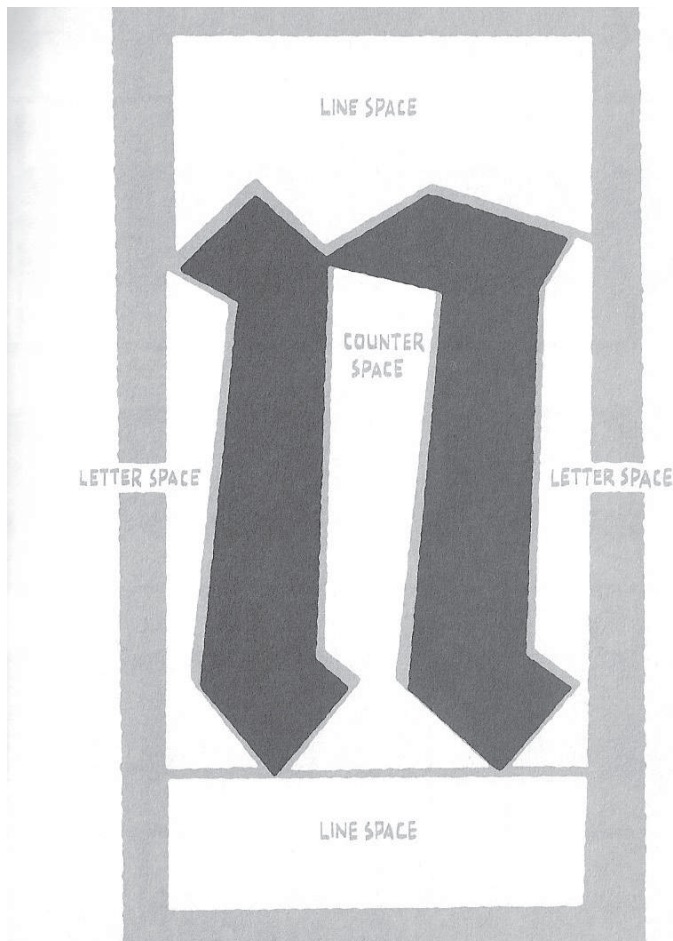
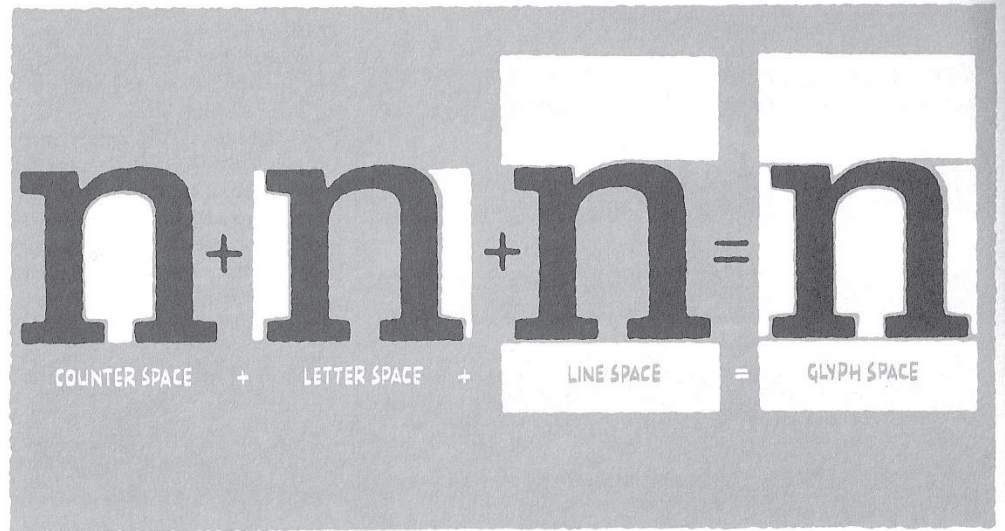
Line space is similar to letter space but it involves vertical relationships instead of horizontal. Between two consecutive lines of text is a white puzzle piece that connects them. A straight horizontal border can be imposed between the lines, splitting the white space so the top half belongs to the letters on the line above and the bottom half belongs to the letters on the line below. There's enough white space to accommodate the descenders of the line above and the ascenders in the line below. These simple borders form the top and bottom sides of the glyph space.

Here's the important result: the glyph space for every letter is the same height, regardless of the height of the actual letter.

DIVIDING THE LINE SPACE SO A BIT BELONGS TO EACH LETTER ABOVE AND BELOW IS KEY TO GUTENBERG'S INVENTION.

Et columnas vi intic
un basibus tot demi
aeneis quaecapitacu
cem caelaturis su se
habebunt age ente as
similiter et in laterial
ilonis per longumeri
tentoriacentumcubi

THE GLYPH SPACE IS FORMED BY ATTACHING THE COUNTER SPACE, LETTER SPACE, AND LINE SPACE TO THE LETTER.



HOW TYPE WORKS

Glyph Space

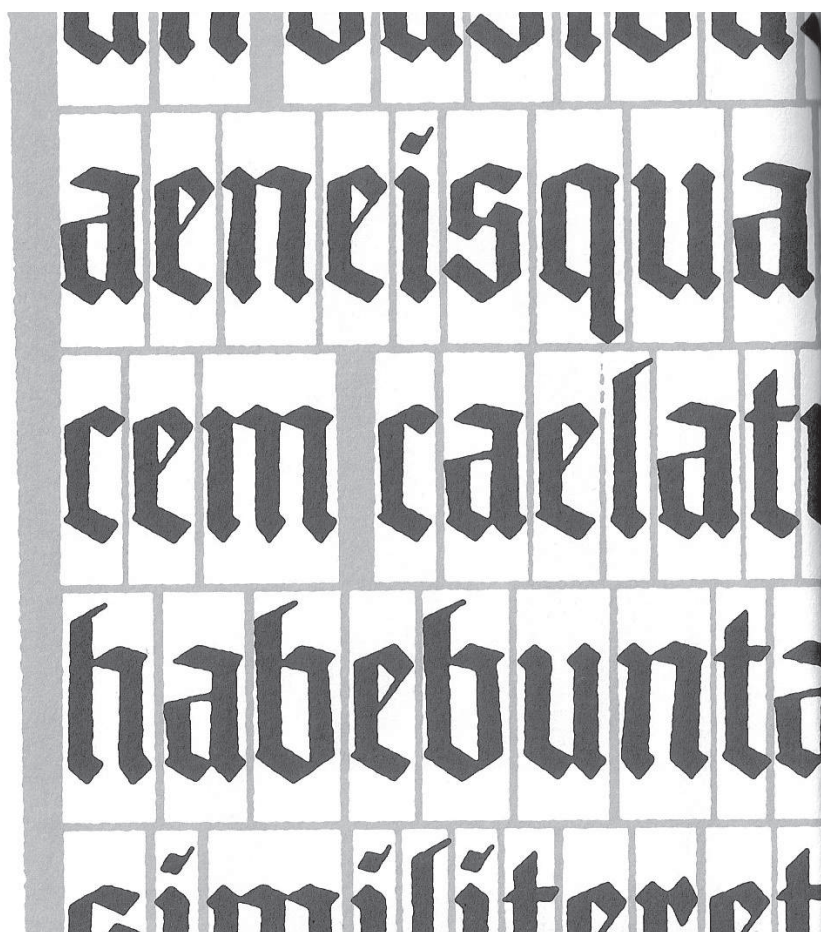
The glyph space is the 'shoebox' of white space surrounding the letter. It's formed by attaching the counter space, letter space and line space to the letter.

You may encounter the term 'body' when referring to the boxes that surround each letter. In fact, you could use the terms 'glyph space' and 'body' more or less interchangeably. When discussing foundry type, 'body' is commonly used and refers to the actual little piece of metal. 'Glyph space' is helpful when discussing the space as a component within a paragraph.

Thinking about white space (or negative space) is not specific to typography or even calligraphy. It's a basic part of two dimensional design. Glyph space, however, is solely a typographic concept. It's the mechanism that makes movable type possible. The glyph space forms a typeface's built-in spacing, and that built-in spacing is the starting point for the structure of a paragraph.

For more on *glyph space within a paragraph*, see page 75.

THE WIDTH OF THE GLYPH SPACE DEPENDS ON THE SHAPE OF THE LETTER BUT THE HEIGHT IS ALWAYS CONSISTENT.



HOW TYPE WORKS

Movable Type

Movable type mimics handwritten text but doesn't reproduce it perfectly. Imposing this kind of strict modularity naturally comes with limits. Gutenberg (and subsequent type designers) worked to adapt calligraphic letters to function within these limits. The typographic letters that they created have evolved into their own species, ranging from geometric sans serifs to elaborate decorative scripts. Digital type has now freed type designers from some of the limits of foundry type, allowing type to evolve further still.

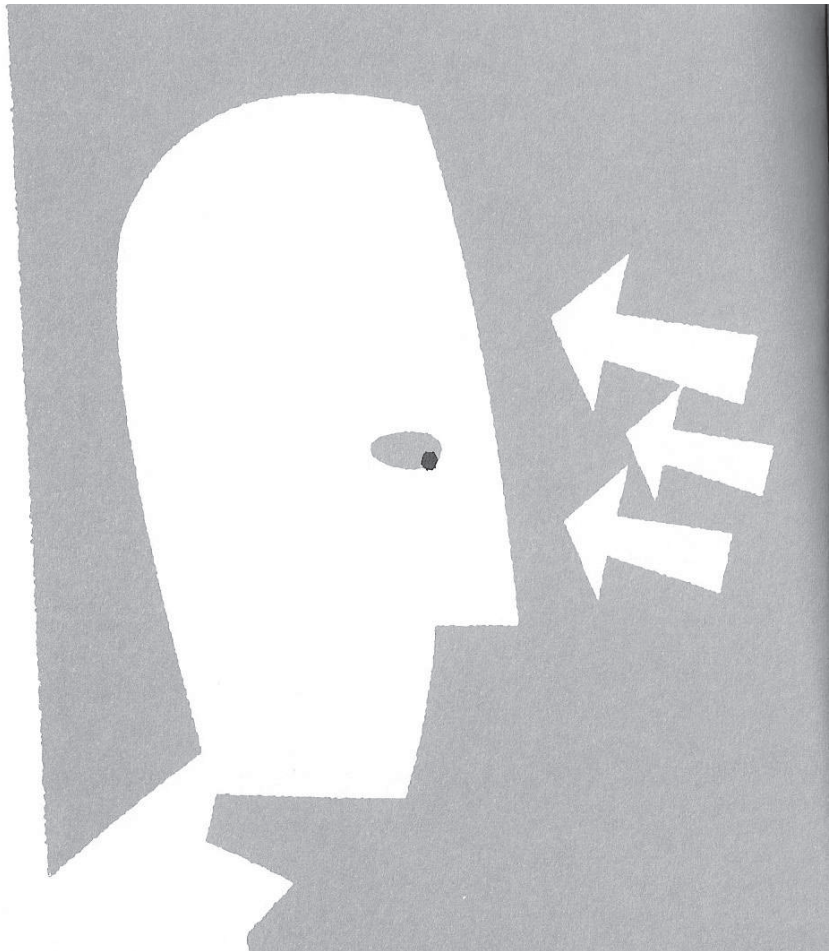
However, the concept at the heart of movable type — the idea of glyph space — is still just as critical to how type works. Thanks to today's software, you can set type without thinking about the concept of glyph space. However, a good understanding of it can help to make you a better typographer. When setting a paragraph, the majority of the typographer's decisions involve inserting or removing space relative to each character's glyph space.

THE GLYPH SPACE FORMS THE INVISIBLE STRUCTURE OF A PARAGRAPH.

how we read

HOW WE READ

A COMMON MAXIM AMONG type designers is that drawing a typeface isn't about drawing beautiful letters: drawing a typeface is about making beautiful words. A typeface is a collection of separate parts, but these parts have to work together in every possible combination to form unified words.

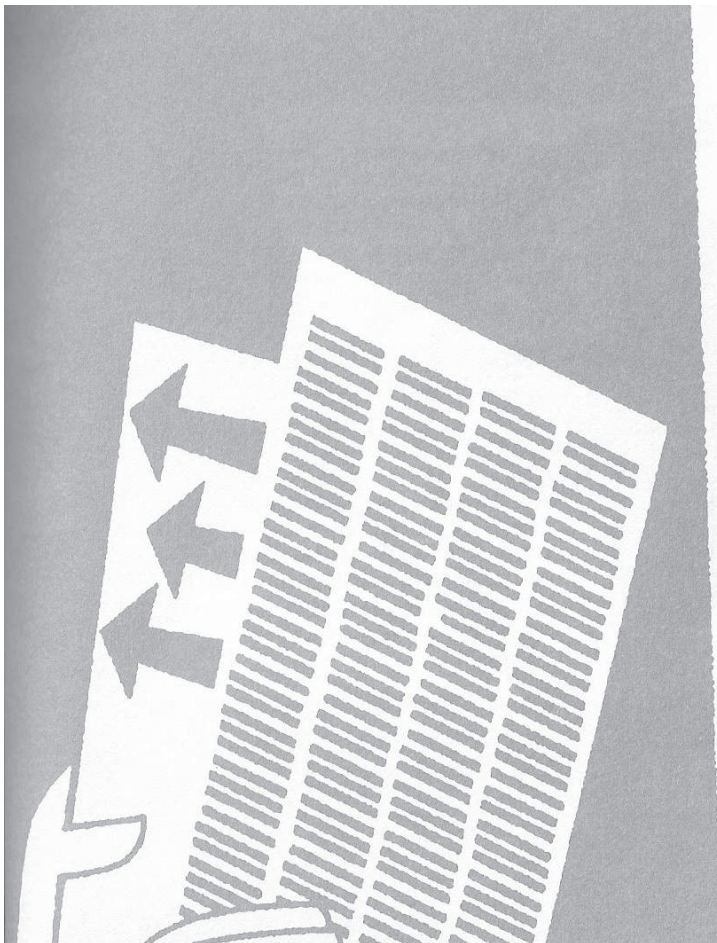


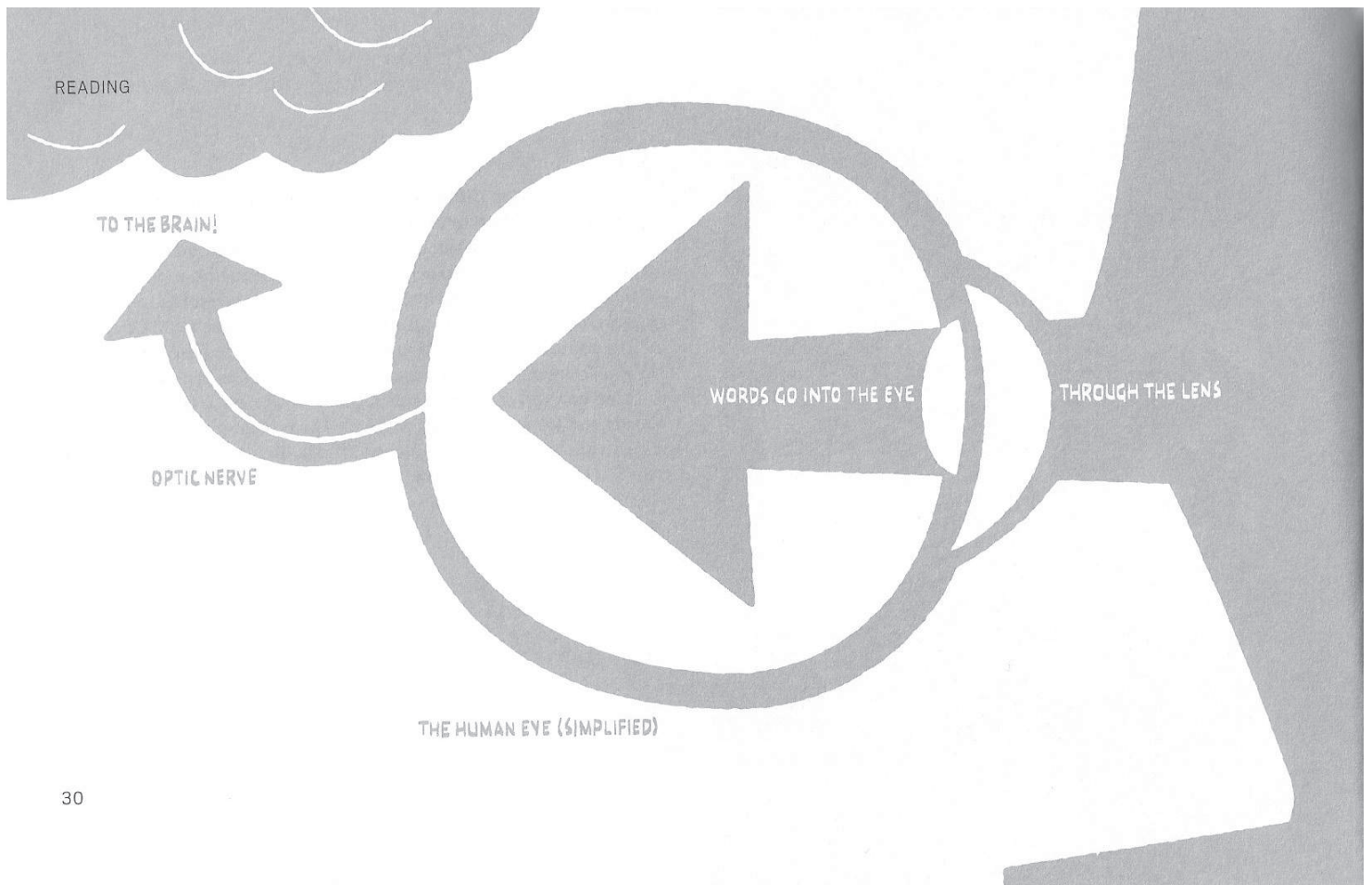
HOW WE READ

Words

Words can take many different forms. A word can be short, even just a single letter, like *a* or *I* in English, *y* in Spanish, or *z* in Polish. A word can be long, like the unbelievably long Welsh place name *Llanfairpwllgwyngyllgogerychwyrndrobwlllantysiliogogoch*. A word can be all lowercase letters, like many of the words in this paragraph, or a word can be all uppercase letters, like *UNESCO*. A word can contain a combination of uppercase and lowercase, like the name *McGrew*. A word can contain punctuation, as in *x-ray*. And a word can contain combinations of these things, as in *O'Connell*.

Words, not letters, are what we read. The typographer can make reading those words easier or more difficult. To learn how to make it easier, it helps to have a basic understanding of how we read.





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READING IS A COMPLEX PHENOMENON. BOTH TYPOGRAPHERS AND SCIENTISTS HAVE STUDIED IT IN ATTEMPT TO FIGURE WHAT EXACTLY IS GOING WITH OUR EYES AND BRAINS WHEN WE ARE READING. OFTEN, THE SCIENTIFIC OBSERVATIONS SEEM TO CONFLICT WITH THE TYPOGRAPHER'S PRACTICAL EXPERIENCE.

ALL CAPS CATCH YOUR EYE BUT NOT MANY READERS WOULD WANT TO READ A WHOLE BOOK SET THAT WAY.

Reading is a complex phenomenon. Both typographers and scientists have studied it in attempt to figure what exactly is going with our eyes and brains when we are reading. Often, the scientific observations seem to conflict with the typographer's practical experience.

OR A BOOK SET LIKE THIS. READERS' EXPECTATIONS AND PREFERENCES ARE IMPORTANT FACTORS IN READING.

HOW WE READ

Reading

Reading is a complex phenomenon. Both typographers and scientists have studied it in an attempt to figure out exactly what's going on in our eyes and brains when we are reading. Often, the scientific observations seem to conflict with the typographer's practical experience.

For example, there are two ways to judge the readability of uppercase text versus lowercase text.

A typographer will most likely tell you that readers don't like uppercase text. A common theory is that this is because uppercase is too monotonous. In contrast, a scientist might tell you that uppercase text is just as readable as lowercase text. In the lab, it can be proven that a reader can become accustomed to reading uppercase, and read it just as easily as they can traditionally-set lowercase.

The problem with the scientific conclusion is that it studies only ability, not expectations or preference. In the lab, readers can even get used to reading text set backwards as easily as text set the normal way. Human guinea pigs are not very likely to put down a page because they don't like the way the type is set; they'll do their best to accomplish their task. However, presented with a newspaper that's set in an unfamiliar way, a reader might well give up and put down the paper.



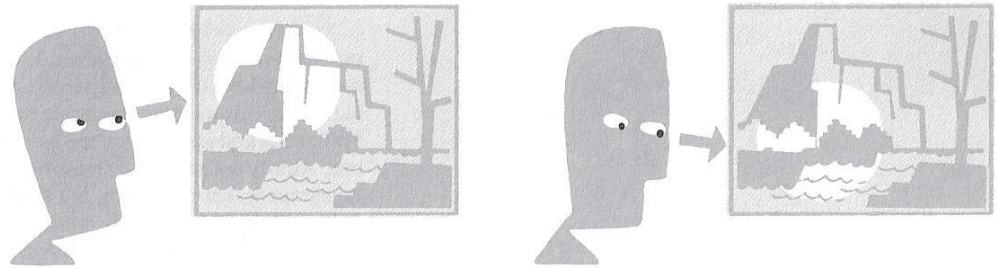
DURING READING, OUR EYES ARE CONSTANTLY JUMPING AROUND IN SMALL IRREGULAR MOVEMENTS CALLED 'SACCADES'.

HOW WE READ

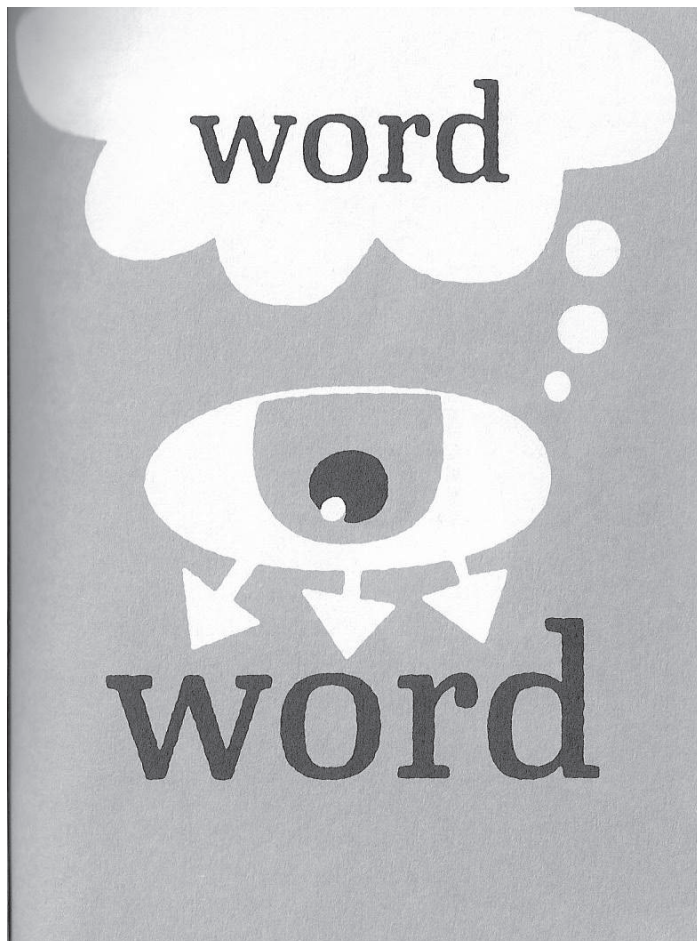
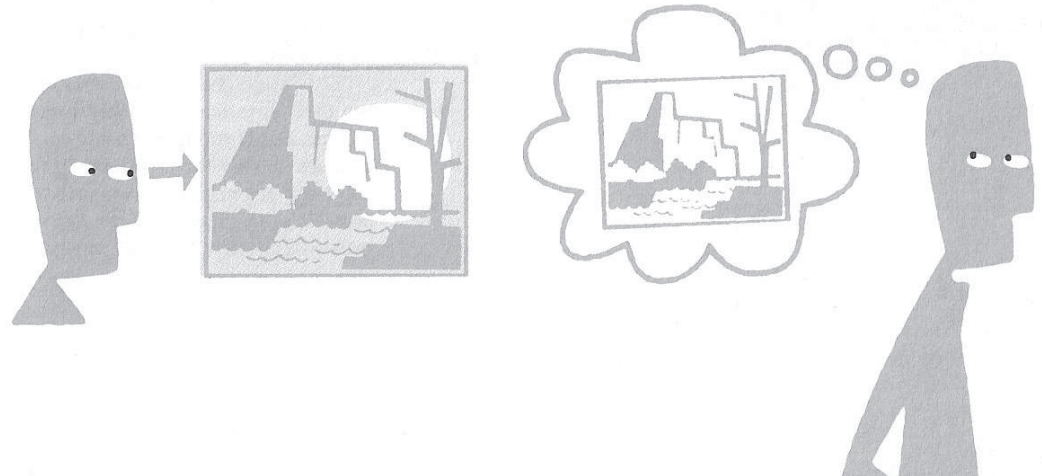
Tradition and reader preference are considerations that typographers take into account when deciding how to compose a document. Readers aren't used to all uppercase text and will find it harder (or just more annoying) to read. They prefer lowercase text. Preference and expectation are hard to measure in a scientific context. Ability (as studied by scientists) and habit (as understood by typographers) are different, but both affect ease of reading.

This doesn't necessarily mean that typographers have it all figured out. Relying only on tradition will give you a limited understanding of typography. The theory that uppercase text is monotonous sounds good and seems to make sense visually, but the scientific conclusions suggest that it's more complicated than that. In the end, it's hard to fit all these different pieces into a cohesive picture of how we read.

Try watching someone's eyes as they read. You might expect to see their eyes scan smoothly across the page, following the lines of type. But if you look closely you might be able to see that their eyes are flicking back and forth in very small movements. You don't need any scientific instruments to conclude that when we read, something is going on that's more complicated than sequentially processing letters from left to right.



WHEN YOU LOOK AT AN IMAGE, YOUR EYE MOVES AROUND SCANNING THE DIFFERENT PARTS. IN YOUR MIND, YOU SEE THE IMAGE AS A WHOLE.



Imagine This

When you look at a drawing, you observe the whole thing. Your eyes move around the image, taking it all in. Different parts might attract your attention first, and the composition will direct your eye around the page, but in your mind, you probably think of the drawing as a whole, single image.

A word is like a drawing. Your eyes move around to take it all in, but your mind perceives it as a whole thing. But reading is a much faster and less random process than looking at a drawing. The eye movements are more practiced and automatic.

In fact, when we read, our eyes are constantly jumping around in discreet irregular movements. Back in the 1880's, a French ophthalmologist named Émile Javal observed this phenomenon and coined the term *saccade* to describe it.

A WORD IS LIKE AN IMAGE, YOUR EYES MOVE AROUND SCANNING THE DIFFERENT PARTS, IN YOUR MIND, THE IMAGE OF THE WORD IS STABLE.

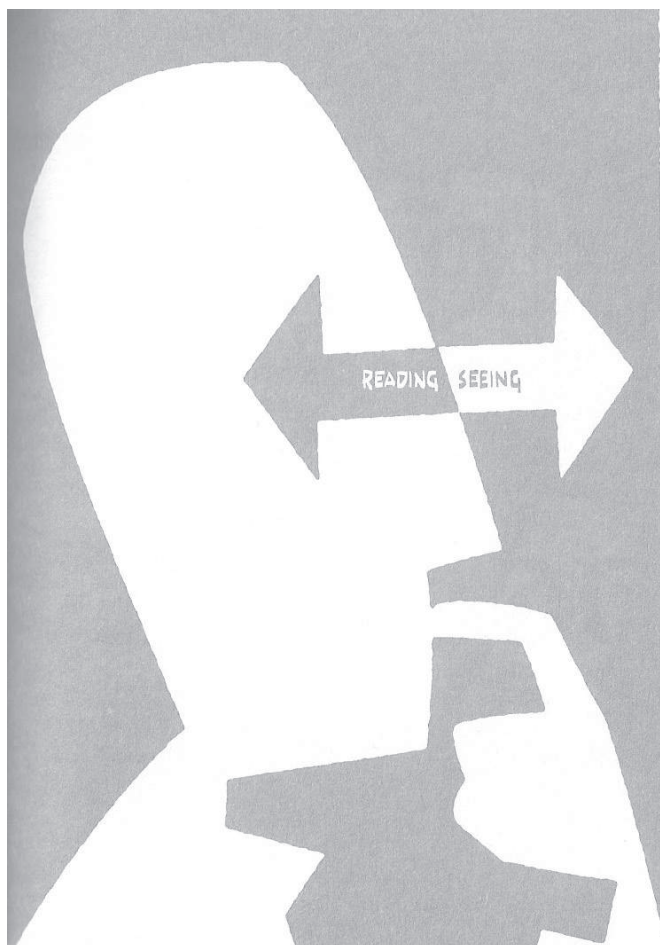
The area of focus

OUR EYES TAKE IN THE WORD
THEY ARE READING...

THEN FLICK AHEAD TO GET AN
IDEA ABOUT THE BEGINNING OF
THE NEXT WORD...

AND SCAN AHEAD FURTHER
TO GAUGE THE LENGTH OF
THE FOLLOWING WORD.

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HOW WE READ

Area of Focus

Using saccades, your eyes take in the word you are reading, while flicking ahead to get an idea about the next word, looking at the first one or two letters, and even flicking further ahead for clues about how long that word is. The sequence of the letters is still important, of course. The eyes quickly flick forward and backward across the word, but the image of the word in your mind is stable.

All this action takes place within a small area of focus. Readers' areas of focus are limited by their eyesight and how much practice they have reading. The job of the typographer is to set the type so that it works well for the intended audience. Two of the biggest factors to consider are the point size and the tracking.

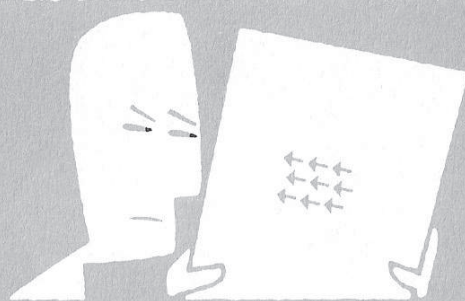
Point size and tracking determine how much is going on inside the reader's area of focus. With experience, the typographer learns how to get the right balance between too much and too little.

THE TYPOGRAPHER LEARNS TO SEE THE TEXT AS WELL AS READ IT.
IN THIS WAY, THEIR EYES RECOGNIZE WHAT LOOKS RIGHT AND WRONG.

37

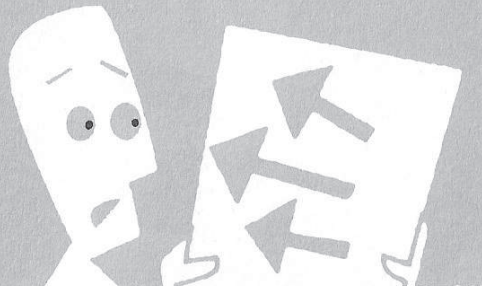
THE RIGHT POINT SIZE FOR TEXT IS INFLUENCED BY THE TYPE OF DOCUMENT AND ITS AUDIENCE.

IF THE TYPE IS TOO SMALL, WE CAN'T DECIPHER THE LETTERS AND THE WORDS CANNOT BE READ.



A COMFORTABLE POINT SIZE FOR TEXT IS USUALLY AROUND 8-10 POINTS.

TYPE THAT'S TOO BIG IS ALSO HARD TO READ. THE EYES CAN BE OVERWHELMED AND HAVE A HARD TIME FOCUSING ON ENOUGH OF THE WORD AT ONCE.



HOW WE READ

Point Size

Obviously, the type shouldn't be set too small. Most readers have had the experience of not being able to read something because the words just weren't big enough to make out. If the type is too small for your eyes, you can't decipher the letters and the words are illegible.

As we age, it gets even harder for our eyes to see small details. Readers with low vision prefer type set at larger point sizes. However, type can also be more cumbersome to read when it's set too large.

When the text is too big, the words don't fit comfortably within the area of focus. It's like looking at a large painting from only one or two steps away. You can't see the whole thing at once. For most text faces, a comfortable size for people with good vision is between eight and ten points.

Chop

10 PTS

TYPOGRAPHERS USE POINT SIZE TO REFER TO THE HEIGHT OF A TYPEFACE'S GLYPH SPACE OR BODY.

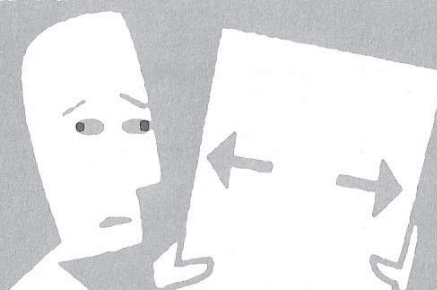
pod pod

10 PTS

POINT SIZE DOESN'T DIRECTLY REFER TO THE SIZE OF THE LETTERS. TO THE READER, TYPEFACES SET AT THE SAME POINT SIZE CAN APPEAR TO BE DIFFERENT SIZES.

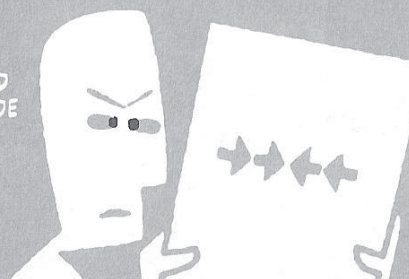
WITH EXPERIENCE, THE TYPO-
GRAPHER LEARNS TO RECOGNIZE
THE RIGHT BALANCE BETWEEN
TOO TIGHT AND TOO LOOSE.

TEXT SET TOO LOOSELY CAN BE HARD
TO READ. THE LETTERS RESIST BEING
CONNECTED INTO WORDS.



THE BUILT-IN SPACING IS ALWAYS A GOOD PLACE TO START FOR BALANCED TRACKING.

TEXT SET TOO TIGHTLY CAN ALSO BE HARD
TO READ. TOO MUCH GETS CRAMMED INSIDE
THE READER'S AREA OF FOCUS.



juniper

THE BUILT-IN SPACING IN A FONT DESIGNED FOR TEXT IS GAUGED TO
APPEAR NEITHER TOO TIGHT NOR TOO LOOSE WHEN SET AT 8-10 PTS.

juniper

LETTER SPACE

COUNTER SPACE

IN A TEXT FACE, THE LETTER SPACE IS USUALLY SLIGHTLY SMALLER
THAN THE COUNTER SPACE.

HOW WE READ

Spacing

If there's too much space between letters, the structural integrity of the word deteriorates. Letters that get spread too far out will fall outside the area of focus. Resolving the letters into words becomes difficult and slows the reader down.

Letters that are too close together can also be hard on the reader. When too many things are crammed into the reader's area of focus, the reader will quickly get overwhelmed and their eyes worn out. The built-in spacing in a text font is usually a good interval. This interval is gauged by the type designer so that the space between the letters is slightly smaller than the space inside the letters.

...spindle-shaped, occasionally phosphorescent, and more rapid in its movements than a whale. To mention such facts relating to this apparition (entered in various most respects as to the shape of the object or the surprising rapidity of its movements, its surprising and the peculiar life with which it seemed endowed surpassed in size all those hitherto classified in scientific consideration the mean of observations made at divers timid estimate of those who assigned to this object a hundred feet, equally with the exaggerated opinions which mile in width and three in length—we might fairly consider it being surpassed greatly all dimensions admitted of ones of the day, if it existed at all. And that it *did* exist, a fact; and, with that tendency which disposes the human mind to the marvellous, we can understand the excitement of the world by this supernatural apparition. As to the question, the idea was out of the question. On the 2nd of March, Governor Higginson, of the Calcutta, met this moving

...Australia. Captain Baker... its exact position when two... fly feet up into the air. Now, und... igginson had to do neither more... columns of water mixed with air... the same year, in the Pacific Ocean... ordinary creature could transport... e Governor Higginson and the Col... an seven hundred nautical leagues... he Helvetia, of the Compagnie... at portion of the Atlantic by... '15° N. lat. and 60° 35' W. lon... length of the

...the past vessels had been met... a long object, spindle-shap... escent, and infinitely lar... ments than a whale

...WHICH AGITATE... LATION AND EXCITED T... N IN THE INTERIOR OF C... AFARING MEN WERE PAR... MERCHANTS, COMMON S... OF VESSELS, SKIPPER, I... AMERICA, NAVAL OFF... AND THE

HOW WE READ

The Typographer's Job

The typographer's job is to work within the limits of readers' preferences and eyesight so that the words can go smoothly into their brains. With experience, the typographer is able to judge what settings will be the most comfortable for the reader. But it also goes further than that. Not only does the typography affect the reader's comprehension, it can affect the reader's perception of the author's voice and ideas.

It's like being able to judge the right volume and speed when you're talking and listening. Some people speak too softly, while others shout. Or you might encounter someone who speaks so fast it's hard to understand them, while others speak too slowly. Usually you can still understand what the speaker is saying, but if it gets to be too much work, you can lose focus and get too distracted to understand what's being said. If the speaker's voice is inappropriate for his message and the context, you can be left with a bad impression of the speaker himself. Type works in a similar way. The typographer sets the text in the appropriate voice — neither too big nor too small, and neither too tight nor too loose. It's possible for readers to decipher poorly-set text, but they might not receive the author's message as well.

letter space

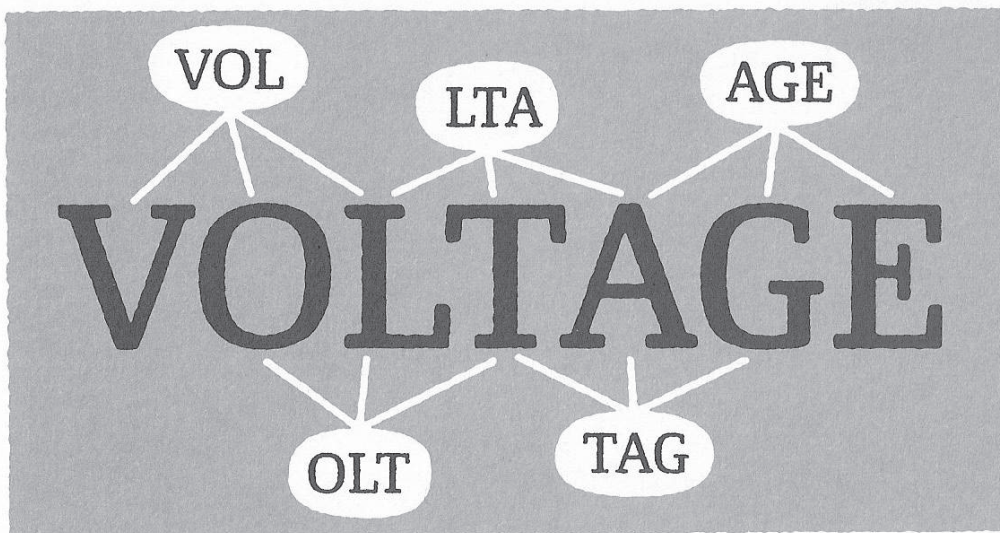
LETTER SPACE

UNDERSTANDING HOW WE READ and how type works gives you a good overall view of how to set text. This next section will be more practical and more detailed. The focus will be on what goes on inside a paragraph — the things typographers see when they look at text. First, letter spacing will be explained, then word spacing and line spacing, and finally there will be a guide for how the typographer can adjust these different kinds of spacing within a paragraph to make it more readable.

FOR EVEN LETTER SPACING, FIND A BALANCE BETWEEN THE LARGE AND SMALL SPACES.

ISOLATE THE LETTERS INTO GROUPS OF THREE.

IN EACH GROUP, THE MIDDLE LETTER SHOULD APPEAR TO BE CENTERED BETWEEN THE FIRST AND LAST LETTER.



VOL
THERE IS TOO MUCH SPACE BETWEEN THE 'VO'.

VOL
NOW THE 'O' IS TOO CLOSE TO THE 'V'.

VOL
THE 'O' APPEARS CENTERED BETWEEN THE 'V' AND 'L'.

Even Spacing

Setting a single word with even letter spacing is a classic training exercise for the beginner. In practice, a good digital typeface should not really need much (if any) manual spacing adjustment within a word. Nonetheless, it's worth reviewing how to compose a word with even letter spacing.

A well composed word should appear to have approximately the same amount of letter space between each letter. Letters shouldn't cluster together at one end and seem afraid of each other at the other end. In a well composed word, the letters appear to be spaced in an even rhythm.

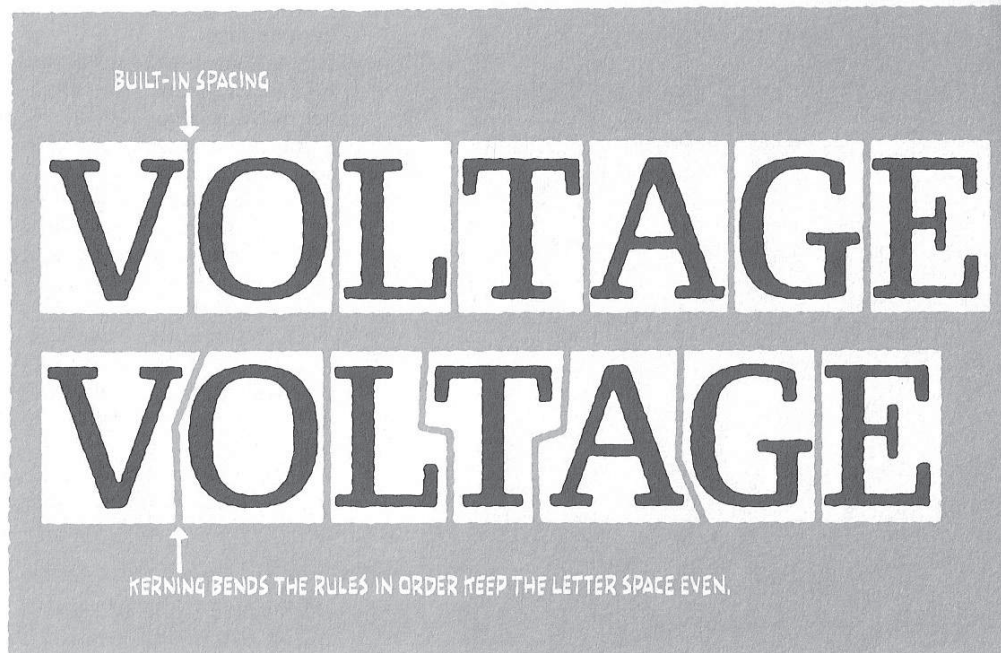
The trick to even letter spacing is to balance the large white spaces between characters like 'TA' that can be eliminated by tucking them together, and the white spaces between characters like 'LA' which cannot be eliminated without smashing the letters into each other.

Type designers take care of most of this when they create a typeface's built-in spacing, but the typographer can fine-tune the letter spacing by adjusting the kerning and the tracking.

USE YOUR EYE TO CENTER THE MIDDLE LETTER.
THIS METHOD WORKS THANKS TO THE OVERLAPPING GROUPS.

THE GLYPH SPACE FORMS A TYPEFACE'S BUILT-IN SPACING.

KERNING IS USED TO ADJUST THE LETTER SPACE BETWEEN SPECIFIC PAIRS OF LETTERS WHEN THE BUILT-IN SPACING DOESN'T WORK.



Kerning

Kerning is a special adjustment to the letter space between a pair of adjacent characters. It's needed because of the modular nature of movable type. The rectangular glyph spaces surrounding the characters limit how they can be spaced. In the majority of character combinations, these limits don't prevent even letter space. However, some combinations are more complicated and the rectangular glyph space gets in the way.

Kerning adjustments, or kerning pairs, are built into a typeface by the type designer. Combinations like 'Ta', 'AV', 'LT' and many others are usually kerned to reduce the white gaps that are formed between these particular pairs of letters. The typesetting software uses a font's built-in kerning pairs to automatically make kerning adjustments as the text is being set.

The typographer can also kern a pair of letters closer together or farther apart from within the document they are setting.

The typographer shouldn't need to make any kerning adjustments when setting a paragraph at text sizes. However, when setting a headline, it's good practice to check the letter spacing using the method described earlier. Any inconsistencies can be solved with manual kerning adjustments.

TRACKING IS A GLOBAL CHANGE
TO THE BUILT-IN SPACING.



Tracking

Tracking is an adjustment to the letter space throughout an entire word, line or paragraph. It adds or subtracts the same amount of space between every character. Tracking adjustments are made by typographers when they're composing documents.

When looking at a document, typographers often notice the overall tightness or looseness of the text.

They might note that the spacing appears loose. Each letter is set far away from the others with a lot of letter space between them.

They might note that the spacing appears tight. Each letter is set very close to the next with very little letter space between them. Letters might even be colliding with each other.

Or they might note that the spacing appears normal and harmonious. The amount of letter space between each letter seems comfortable for reading — not too loose and not too tight. In text, this usually means the letter space is slightly smaller than the counter space. If the overall spacing needs correcting, the typographer can adjust the tracking.

The optimum amount of spacing depends on the point size of the text. Get to know the typeface you're using. Depending on what it was designed for, its built-in spacing will be tuned in different ways. A headline face will be spaced tighter by default than a series designed for text. The typographer needs to adjust the tracking accordingly — it doesn't happen automatically.

Usually, the optimum amount of letter space for any point size appears to the reader as neither too tight nor too loose. Typographers use their judgement to adjust the tracking in order to get the right amount of letter space.

Here are some general guidelines for good spacing:

Headlines should have slightly tighter letter spacing.

Text should have normal letter spacing.

Fine print should have slightly looser letter spacing.

All-uppercase settings should be set with positive tracking.

TAKE A MOMENT TO EVALUATE THE BUILT-IN SPACING AS YOU CHANGE THE POINT SIZE.



DIGITAL TYPEFACES CAN BE SET AT ANY POINT SIZE THE TYPOGRAPHER CHOOSES. AS THE LETTERS GET SCALED UP OR DOWN, THE GLYPH SPACE SCALES ALONG WITH THE LETTERS. THE RESULT IS THAT A TYPEFACE MAY APPEAR LOOSE AT BIG POINT SIZES BUT TIGHT AT SMALL SIZES.



THE TYPOGRAPHER CAN ADJUST FOR THIS USING TRACKING.

LETTER SPACE

Headlines & Text

Headlines should catch the reader's eye. The words should be compact so the eye can take them in quickly. If the letter space is too loose, the reader will take longer to resolve the letters into words. When setting a headline, take just a moment to evaluate the letter space. You might find it looks better with slightly tighter tracking. In your typesetting software this usually means setting the tracking value to a negative number.

Text is different. Our eyes have less trouble resolving the letters into words because they're smaller and fit more easily into our area of focus. The letters should have enough breathing room to distinguish themselves without having so much tracking that the integrity of each word is compromised.

TRADITIONALLY, TYPOGRAPHERS
ADD TRACKING WHEN SETTING
WORDS IN ALL UPPERCASE.

HELIUM

THE UPPERCASE ARE SPACED TO GO WITH THE LOWERCASE, NOT EACH OTHER.
THEIR BUILT-IN SPACING IS TOO TIGHT FOR ALL CAP SETTINGS.

HELIUM

INCREASE THE TRACKING UNTIL THE LETTER SPACE IS JUST SLIGHTLY SMALLER
THAN THE COUNTER SPACE.

LETTER SPACE

Fine Print & All Caps

Fine print (or small text) is the hardest to read. At the smallest sizes, letters can appear to run together and be hard to distinguish. To compensate for this, small text should be subtly tracked out so that the letters have plenty of space around them and they can be identified. When setting small text, you can loosen the tracking by setting the tracking value to a positive number.

Finally, it's a typographic convention for the typographer to add some positive tracking to text set in all caps. In a typical typeface, the built-in spacing favors the lowercase. The wide uppercase is spaced to work well with the narrower lowercase rather than with other uppercase. Therefore, all-uppercase letter combinations can seem tight. Recently, thanks to software advances, the font's built-in spacing can compensate for this and automatically add tracking between uppercase characters. So typographers need to use their judgement about whether or not to add positive tracking to an all-caps setting.

text set smaller
than eight points
will usually need
extra letter space

hurry
S L O W D

up
O W N

LETTER SPACE

Small Moves

In all of these examples, the recommended tracking adjustments are made within the limits of what the reader will perceive as normal letter spacing. They're usually small moves, not big ones.

When typographers adjust tracking, they're changing the built-in spacing put into the typeface by the type designer. It isn't against the rules to do this, but the typographer should have a good reason. Digital typesetting has made this kind of adjustment so painless, it's easy to do without thinking. But the balance between the counter space and the letter space is delicate. Upsetting this balance can negatively impact readability.

The typographer can, of course, make even bigger changes to the letter spacing for dramatic effect.

word space & line space

WORD SPACE & LINE SPACE

IN ADDITION to controlling the space between the letters, the typographer can control the space between the words, and the spaces between the lines themselves. The typographer's job is to adjust the size of these spaces to create a well-defined line of text that guides the reader's eye across the page, connecting the words into thoughts. Then, as the reader moves from one line to the next, the thoughts are connected into fuller ideas.

WWW.SCRIPTURACONTINUA.COM

THE ANCIENT TRADITION OF NO SPACE BETWEEN WORDS IS ALIVE TODAY IN THINGS LIKE WEB ADDRESSES, HOWEVER, MOST MODERN READERS USUALLY PREFER A DISTINCT SPACE BETWEEN THE WORDS.

WORD SPACE & LINE SPACE

Word Space

Putting space between words wasn't always done. In the ancient Roman world, writing was done in *scriptura continua*. One word just bled into the next one, without any space in between. As a modern reader you'll find that this will slow you down, but you can still read text without word spaces if you try. Think of the URLs we decipher every day. Sometimes we get stuck momentarily on something new, but we can figure it out. However, reading a whole paragraph without any word spaces is more work than most readers are willing to put in.

Word spaces help us read more fluidly, and appropriately sized word spaces make reading even easier. If the word space is too small and the words bleed together, the text can be deciphered, but reading will slow down. Word space that's too big can make it difficult to string the words together into complete thoughts. As with all the kinds of spacing inside a paragraph, there's a balance between too much and too little.



THE WORD SPACE CAN BE ADJUSTED INDEPENDENTLY OF THE TRACKING.

word space

THE RIGHT SIZED WORD SPACE IS A BALANCE BETWEEN SEPARATING AND CONNECTING WORDS.

word space

IF IT'S TOO BIG, THE WORD SPACE SLOWS DOWN THE READER AND THE FLOW OF IDEAS.

The facts relating to this apparition agreed in most respects as to the shape of the object or creature in question, the untiring rapidity of its movements, its surprising power of locomotion, and the peculiar life

IN JUSTIFIED TEXT, THE WORD SPACE IS SMALL IN SOME LINES AND BIG IN OTHERS.

WORD SPACE & LINE SPACE

When setting type, the typographer can change the word space independently from the rest of the font's built-in spacing. Unlike tracking, which should be adjusted cautiously in small increments, word space can absorb bigger changes since it's a relatively large space within the paragraph. Of course, the typographer still needs to be careful with the word space, but it's generally less delicate than letter space.

In a column of text, the typographer usually makes any word space adjustments to the whole thing at once. For example, a wide column may have a larger word space to give the words some breathing room. A narrow column may have a smaller word space to make the lines more compact. In a column of justified text, the word space can also be adjusted line by line. You've probably noticed this variation in word space in the narrow justified columns of a newspaper. Suddenly, there's a line with big gaps in it. In unjustified text, the word space stays consistent throughout the column. This will be more fully explained in the section on hyphenation and justification.

THE IMPORTANT THING IS HOW THE SIZE OF THE WORD SPACE RELATES TO THE SIZE OF THE LINE SPACE.

LEADING IS THE VERTICAL
DISTANCE BETWEEN BASELINES
OF TWO LINES OF TYPE. IT IS
USUALLY MEASURED IN POINTS.



Line Space

The right-sized word space helps readers connect a line of words into thoughts. Connecting those thoughts into more complete ideas is a matter of getting the reader smoothly from one line to the next. The amount of space between the lines, or line space, has a large effect on this transition.

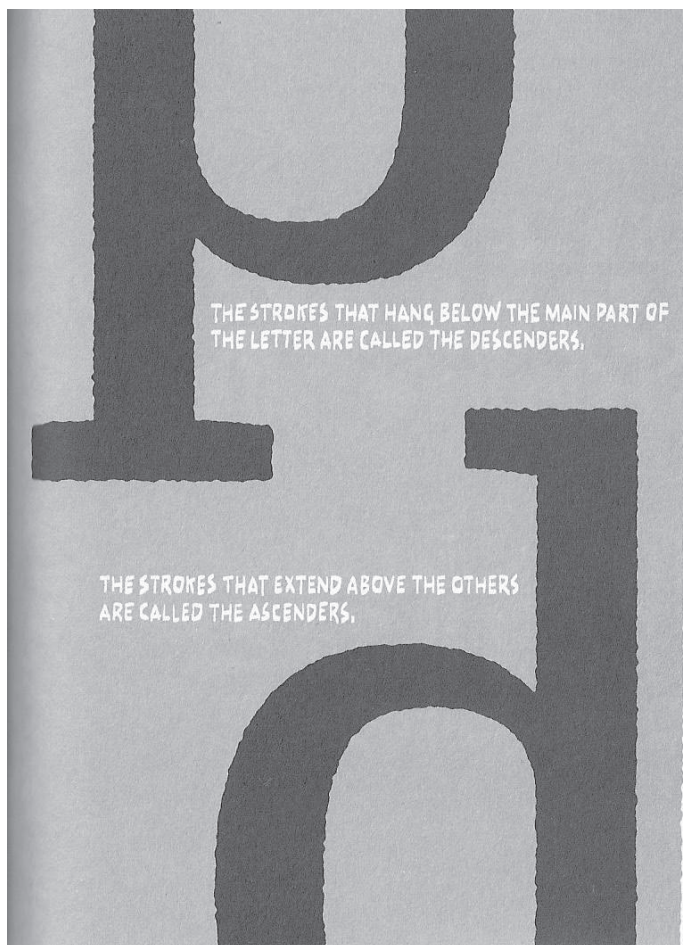
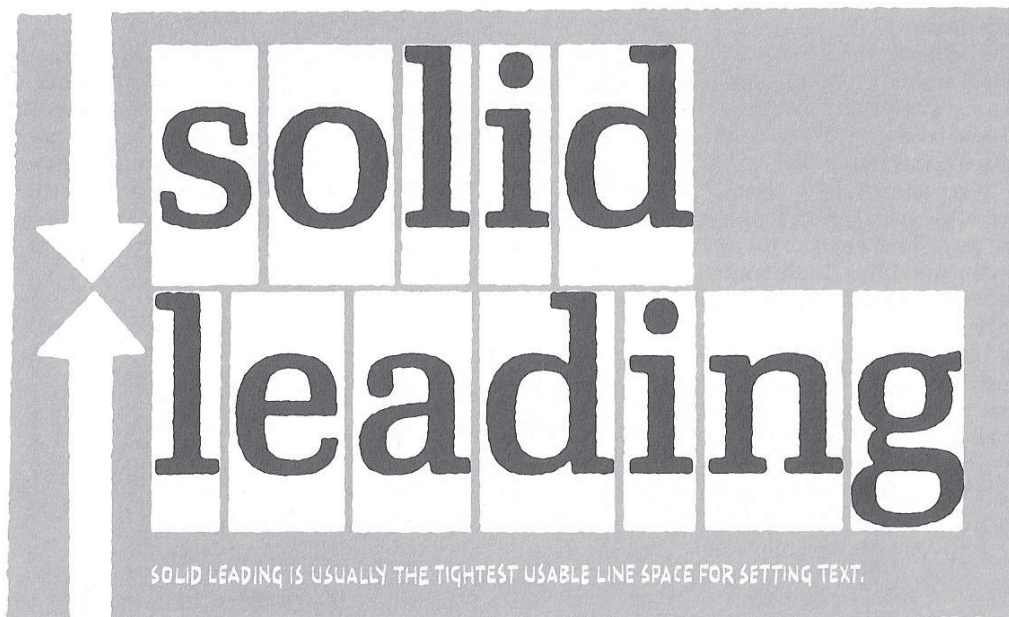
When typographers make adjustments to a font's built-in line space, they're said to be adjusting the leading. The amount of leading used to refer only to the extra strips of lead inserted between lines of metal type. The definition has evolved to mean the distance between the baselines, the imaginary lines that the words appear to rest on.

Leading is measured in points, the same unit system used for specifying type size. The standard practice is to specify point size and leading together. For example, 10-point type with 12-point leading would be specified as 10/12. This is pronounced 'ten on twelve'. Conventionally, the lines within a paragraph have a consistent amount of leading.

A font's built-in metrics for letter spacing are quite sensitive, but the metrics for line spacing are a bit more amenable to change. The built-in vertical metrics usually provide enough line space to prevent collisions between the descenders from one line and the ascenders in the next line. It's expected that typographers will adjust the amount of line space by adding or subtracting leading to suit the particular text they're setting.

THE TERM LEADING IS LEFT OVER FROM THE DAYS WHEN TYPE SETTING
WAS DONE WITH METAL.

HERE, THE POINT SIZE AND LEADING ARE EQUAL, SO THE LINE SPACE REFLECTS THE FONT'S BUILT-IN VERTICAL METRICS.



WORD SPACE & LINE SPACE

Solid Leading

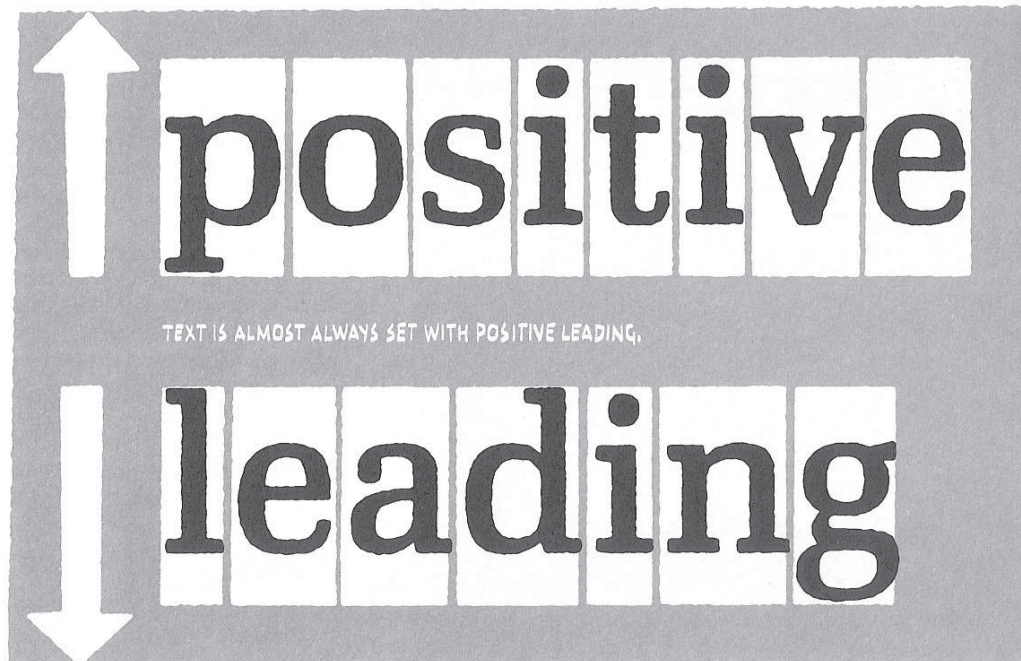
To set the line space according to the font's built-in spacing, set the leading to the same value as the point size (10/10). This is known as a solid setting or solid leading. Solid leading is usually the smallest amount of leading the typographer can safely use in body text. The overall appearance will often be quite cramped. If the line space is any tighter than this, ascender and descender collisions are probably imminent.

In metal type, the tightest line space you could have was a solid setting. In digital type, we don't have this limitation. The leading can be set to a value less than the point size, such as 10/8. This means 10-point type with only 8 points between the baselines. The bodies are actually overlapping and the ascenders and descenders are probably crashing into each other. When setting body text, it's rare to use negative leading. It happens more often when setting compact headlines of two lines or more, especially when using all caps, which usually have no ascenders (although the Q and J do sometimes have small descenders).

IF THE LEADING IS LESS THAN THE POINT SIZE, ASCENDERS AND DESCENDERS MAY COLLIDE.

HERE THE LEADING IS GREATER THAN THE POINT SIZE.

TEXT IS ALMOST ALWAYS SET WITH POSITIVE LEADING.



WORD SPACE & LINE SPACE

Positive Leading

Traditionally, a typographer would add a bit of leading to increase the space between the lines. This is known as positive leading. Most typesetting applications now set type with positive leading by default. With many languages, a good rule of thumb is to set the leading to 120% of the point size. For 10-point type, that would mean 12-point leading, or 10/12.

However, some languages need more leading in order to avoid collisions. A language like Slovak is a good example. Slovakian words contain a lot of diacritic marks that float above the letters. In a digital typeface, the diacritics placed above the uppercase often exceed the height of the body. Extra leading is needed to avoid collisions.

SOME SLOVAKIAN WORDS ILLUSTRATE HOW LANGUAGE CAN INFLUENCE LEADING. THE DIACRITIC MARKS ABOVE THE UPPERCASE CRASH INTO THE DESCENDERS FROM THE LINE ABOVE.



The facts relating to this apparition (entered in various log-books) agreed in most respects as to the shape of the object or creature in question, the untiring rapidity of its movements, its surprising power of locomotion, and the peculiar life with which it seemed endowed. If it was a whale, it surpassed in size all those hitherto classified in science. Taking into consideration the mean of observations made at divers times—rejecting the timid estimate of those who assigned to this object a length of two hundred feet, equally with the exaggerated opinions which set it down as a mile in width and three in length—we might fairly conclude that this mysterious being

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THE TYPOGRAPHER LEARNS THE BALANCE BETWEEN TOO MUCH AND NOT ENOUGH LINE SPACE.

TOO MUCH LINE SPACE DAZZLES THE EYES WITH VIBRANT WHITE STRIPES.

NOT ENOUGH LINE SPACE CREATES DISTRACTING WORD SPACES.

The facts relating to this apparition (entered in various log-books) agreed in most respects as to the shape of the object or creature in question, the untiring rapidity of its movements, its surprising power of locomotion, and the peculiar life with which it seemed endowed. If it was a whale, it surpassed in size all those hitherto classified in science. Taking into consideration the mean of observations made at divers times—rejecting the timid estimate of those who assigned to this object a length of two hundred feet, equally with the exaggerated opinions which set it down as a mile in width and three in length—we might fairly conclude that this mysterious being surpassed greatly all dimensions admitted by the learned ones of the day, if it existed at all. And that it did exist was an undeniable fact; and, with that tendency which disposes the human mind in favour of the marvellous, we can understand the excitement produced in the entire world by this supernatural apparition. As to classing it in the list of fables, the idea was out of the question.

THE RIGHT AMOUNT OF LINE SPACE UNITES THE WORDS INTO WELL-DEFINED LINES.

Line Space & Reading

Avoiding collisions between lines is the bare minimum you should hope to achieve when setting the line spacing. Line spacing also affects the readability of a paragraph.

Too much leading can make it difficult to connect the thoughts in one line to the thoughts in the next. In poetry, this might be a good effect. Line breaks are an intentional part of the writing, and each line should be considered before moving on to the next. In prose, extra line space will make it hard for the reader to follow the author's ideas.

Not enough leading can make it difficult for the reader to keep track of which line they're on. In the middle of a line, the reader's eyes can slip down onto the line below. The reader might also get lost when traveling back from the end of one line to the beginning of the next.

In a narrow column, where the eye doesn't have to travel back very far, less leading is needed. In wider columns, a little additional space between the lines can make it easier for readers to keep track of where they are.

paragraph settings

PARAGRAPH SPACING

WE COULD STOP NOW and become pretty good typographers. If you understand how type works, how we read, and how to see what's going on inside a paragraph, you should be able to compose text without falling into the traps that waylay most novices. In the next section, we'll go a step further by looking at the entire paragraph as a whole. The key to setting text really well, instead of just adequately, comes from understanding the relationships within a paragraph. This is how the experienced typographer puts this basic knowledge into practice.

LETTER SPACE IS THE SMALLEST SPACE WITHIN A PARAGRAPH. LINE SPACE IS THE BIGGEST SPACE WITHIN A PARAGRAPH. THE WORD SPACE IS IN BETWEEN.

TINY LETTER SPACE MEDIUM WORD SPACE BIG LINE SPACE

letter space,
word space,
and line space

The facts re
this apparit
tered in var
books) agre
respects as
shape of th
or creature
tion, the un
pidity of it
ments, its
power of lo
and the pe
with which
endowed.

PARAGRAPH SPACING

Hierarchy of Space

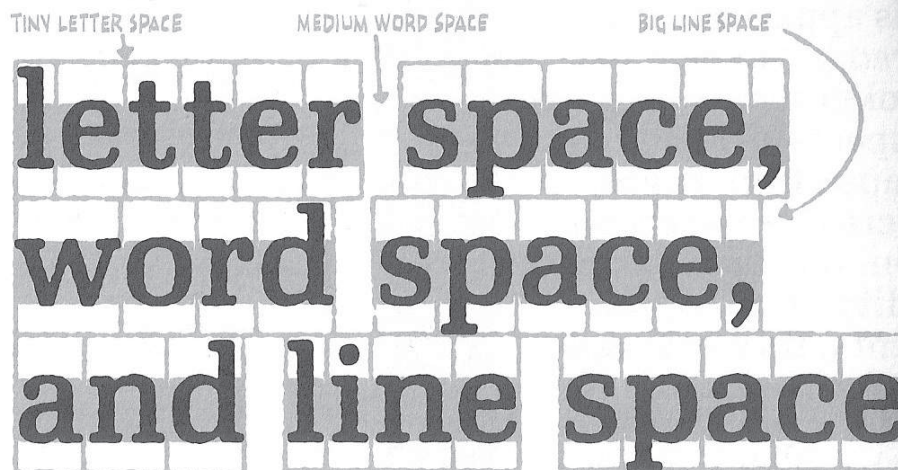
The starting point for the spacing within a paragraph is determined by the type designer, with the font's built-in spacing. The glyph spaces around each part of the typeface form blocks that make up the invisible structure of the paragraph. The typographer's job consists largely of adjusting the size of white spaces — adding and subtracting distance between those blocks. These adjustments affect the letter space, word space, and line space.

With each kind of white space, there's a delicate balance between too much and too little. When you look at them separately, it can be hard to understand where that balance lies. But in the context of a paragraph, it can be easier to see what's going on. The different kinds of white space within a paragraph don't exist in isolation. They have a hierarchical relationship based on size.

Letter space is the smallest, line space is the biggest, and word space is somewhere in between. This hierarchy of white space within the paragraph must be maintained. When it gets disturbed, the parts of the paragraph get muddled and become harder for the reader to navigate.

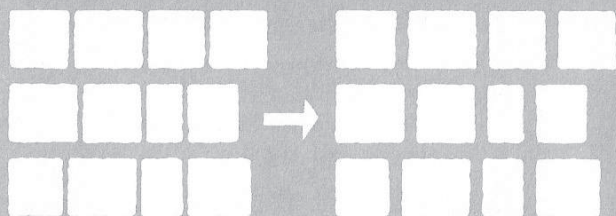
THE SPATIAL HIERARCHY IS EASIER TO SEE IF WE USE SIMPLE BLOCKS TO REPRESENT THE LETTERS.

THE BASIC STRUCTURE FORMED BY THE FONT'S BUILT-IN SPACING HAS THE CORRECT HEIRARCHY OF LETTER SPACE, WORD SPACE, AND LINE SPACE.

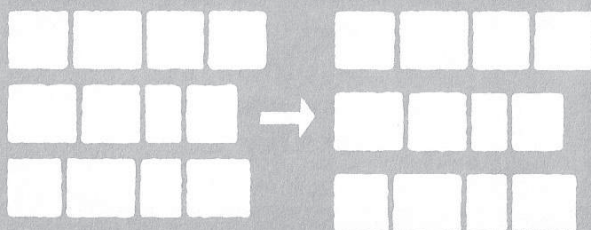


THE GLYPH SPACE MADE VISIBLE REVEALS THE PARAGRAPH'S STRUCTURE.

PARAGRAPH SPACING



CHANGING A SMALL SPACE, LIKE LETTER SPACE, MAKES A BIG IMPACT ON THE APPEARANCE OF A PARAGRAPH.



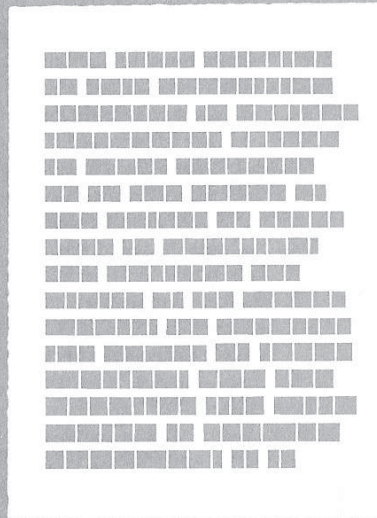
THE SAME AMOUNT OF CHANGE TO THE LINE SPACE, A BIG SPACE, IS LESS NOTICEABLE.

In addition to preserving this size hierarchy, the typographer needs to be aware that the different white spaces have different amounts of importance for the reader. The smaller the white space, the more important it is to the overall readability of a paragraph. Therefore, experienced typographers are very careful when changing the letter space (the smallest space within the paragraph), but they have more flexibility when changing the line space, which is the largest space.

Fortunately, thanks to the way type works, a paragraph set in a well-designed text typeface will have the correct spacing hierarchy by default. The invisible structures formed by the glyph space automatically result in appropriate letter space, word space, and line space. Then it's up to the typographer to carefully adjust these spaces to suit the context and content of the document being composed.

THE TYPOGRAPHER FINE-TUNES THE PARAGRAPH SETTINGS BY CAREFULLY ADDING AND SUBTRACTING SPACE AROUND A TYPEFACE'S BUILT-IN SPACING.

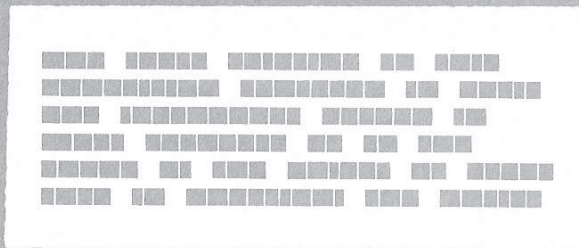
LETTER SPACE, WORD SPACE, AND LINE SPACE WORK TOGETHER TO CREATE AN EVEN TEMPO.



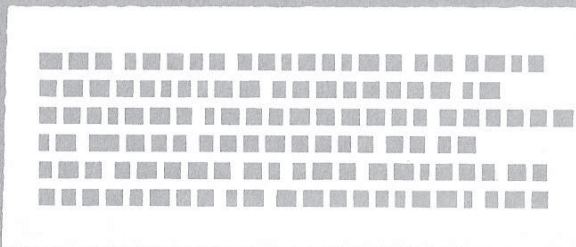
TEXT SET WITH TIGHT SPACING HAS A FASTER TEMPO.



TEXT SET WITH LOOSE SPACING HAS A SLOWER TEMPO.



TIGHT LETTER SPACE WITH A LOOSE WORD SPACE CREATES A CHOPPY TEMPO AND BIG DISTRACTING WHITE GAPS.



LOOSE LETTER SPACE WITH A TIGHT WORD SPACE CREATES A WOBBLY TEMPO AND LINES OF TEXT THAT ARE HARD TO FOLLOW.

PARAGRAPH SPACING

Tempo

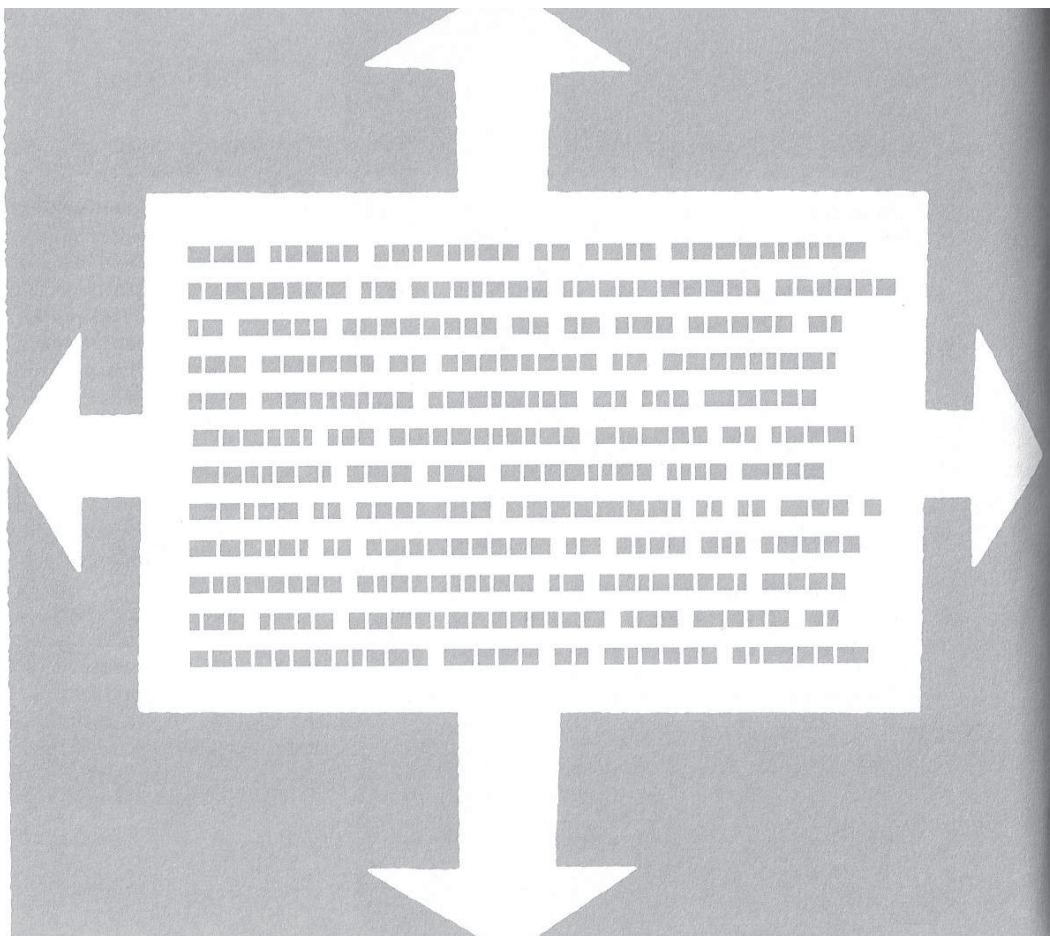
Adjusting the paragraph settings while preserving the spacing hierarchy can be managed in large part by thinking of the pace or tempo of the paragraph as a whole. Each part of the paragraph should have spacing that's compatible with the other parts. This lets the reader move smoothly and easily through the text. Tighter spacing equals a fast tempo while looser spacing results in a slow tempo. To maintain an even flow throughout the paragraph, the tempo should be consistent in each part. For example, tight letter spacing is compatible with tight word spacing. The fast tempo between the letters is supported by the fast tempo between the words.

A line with tight letter spacing but loose word spacing has incompatible kinds of spacing. The words are fast but the space between them is slow. The rhythm of the line is choppy and bad for the flow of text. It's like listening to someone who talks by blurting out one word at a time.

The opposite — loose letter spacing combined with tight word spacing — is even worse. The words are slow but the space between them is fast. The inconsistent tempos are working against each other, and the effect is like a slow monotone voice that never pauses for a breath. No one wants to listen to or read that. It also harms legibility: it becomes hard to see where words begin and end.

COLUMN WIDTH

SPACING SHOULD BE EXPANDED
AS THE COLUMN GETS WIDER AND
CONTRACTED AS THE COLUMN
GETS NARROWER.



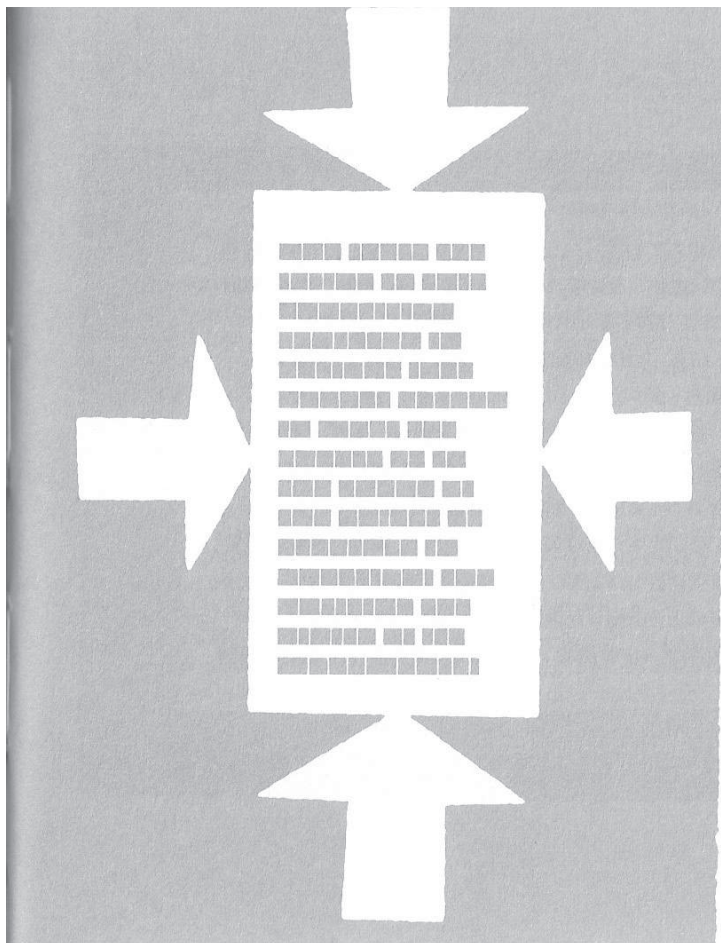
80

PARAGRAPH SPACING

Column Width

The typographer's decision about how to set the tempo of a paragraph is not entirely arbitrary. The appropriate tempo is related to the paragraph's width. As a paragraph gets wider, the tempo should slow down. The white spaces between letters, words and lines should expand. As the column contracts and gets narrower, the tempo should speed up, and the white spaces contract.

But different kinds of spacing change at different rates. Small spaces like the letter space can be adjusted by adding or subtracting a very small amount of tracking. The larger line space can be adjusted by greater amounts. The important thing is to think of the different kinds of spaces as interdependent. As one kind of space expands or contracts, it has an effect on the other kinds of space, and they may need a similar (but not identical) adjustment.



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THIS WIDE COLUMN HAS A SLOW,
RELAXED TEMPO,

THE LETTER SPACE, WORD SPACE, AND LINE SPACE ARE LOOSE, SO THE FLOW OF THE
TEXT IS EVEN AND SMOOTH.

THE LETTER SPACE, WORD SPACE, AND LINE SPACE ARE LOOSE, SO THE FLOW OF THE
TEXT IS EVEN AND SMOOTH.

Wide Columns

A reader might sit in a comfortable chair and read a novel for a few hours at a time. The book's typography should be composed so the reader can move through the text at a leisurely, comfortable pace.

Novels are traditionally set with one wide column of text on a page, compared to the narrower columns you find in newspapers or magazines. Wide columns allow the author to fit a more complete thought onto a line. This helps the flow of ideas go more smoothly, making it easier for the reader to enjoy the text.

The longer the line, the more open the different kinds of space should be. The font's point size should be set so that about 50 to 70 characters fit on a line, including the word space. In English, this equals about 10 to 15 words per line. If you're using a typical serif text face, the point size will probably be about 10 points. Books come in different sizes, of course, but the page margins are usually adjusted along these guidelines when possible.

A good text face is designed so that each letter has a bit of breathing room when it's set between 8 and 10 points. Typographers get to know the typefaces they use so they can judge whether any tracking adjustments are needed. The tempo of the letter space sets the tone for the reader. If it's too fast or too slow, the reader will have a harder time.

Tracking adjustments affect the word space in addition to the letter space. The typographer can also adjust the word space independently. The word space should be set so that each word is distinct and easy to comprehend. It should be compatible with the tempo of the letter spacing. Since the word space is larger than the letter space, it's less sensitive and the typographer has a little more leeway when it comes to adjusting it. If the letter space is loose, an additional adjustment to loosen the word space might be called for.

Line space is the largest and least sensitive white space in the paragraph. It's gauged to prevent readers' eyes from slipping to another line or losing their place on the way back from the end of one line to the beginning of the next. It should be set so that the size of space between the lines is clearly distinguishable from the word space and so that each line is distinct. In a wide column, the text is almost always set with positive leading.

In other words, the longer the line, the more open all the spacing settings should be. Wide columns, designed for longer texts, are spaced so that each element in the paragraph has room to breathe and there isn't any clutter. The tempo is slow and steady. Books are usually designed so the reader can settle in and become engrossed in a long story.

THESE NARROW COLUMNS ARE
SET WITH A QUICKER TEMPO.



THE LETTER SPACE, WORD SPACE, AND LINE SPACE ARE TIGHT, SO THE FLOW OF THE
TEXT IS FAST AND THE LINES ARE COMPACT.

Narrow Columns

At the other extreme, newspapers are designed for shorter texts, meant to be read quickly or even just skimmed while other things on the page are competing for the reader's attention.

Narrow columns are traditionally used in newspapers because they're easy to scan and good for packing a lot of information into a limited amount of space. The point size in a narrow column should be set to fit four to eight words per line (for English this amounts to about 25 to 35 characters). If the point size is too big, not enough characters will fit on a line and the flow of the text will get choppy. The smaller you can set the text, the better — within reason.

The letter space in a narrow column should be compact so that the words are bold and catch your eye. Typefaces specifically designed for newspapers usually have tighter built-in spacing compared to typefaces designed for book work. Typographers should be aware of what kind of document a typeface was designed for. You can also judge by eye whether the tracking needs a small adjustment or not.

The tempo in a narrow column is fast. The word space should be set as tight as possible so the lines are dense and can be skimmed quickly. It's often helpful to establish a reference point by setting the word space way too tight, then slowly making it bigger, until it's just readable enough.

The line space should also be tight, following the spacing of the rest of the paragraph. Tight line space makes it easy for the reader to skim the page and get to the next line. If there's too much line space, the white space between the lines can become a visual distraction and make it difficult to focus on the text.

The goal with all these tight settings is not to jam as much text as possible into a narrow column. The text still needs to be readable. The goal is for the reader to be able to skim through the text quickly before they're distracted by an advertisement or have to get off the train.

fine tuning

FINE TUNING

AFTER THE TYPOGRAPHER adjusts the paragraph settings so that the tempo is appropriate to the column width and the document, the settings can be fine-tuned with the hyphenation and justification settings, or H&Js. The H&Js help determine how the lines of a paragraph should break and what happens when they do.

If the paragraph settings are well adjusted to the column width, setting the H&Js should be fairly easy. In reality, the typographer often goes back and forth between adjusting the paragraph settings and fine tuning the H&Js. Narrow columns typically give the typographer the most trouble and can require a lot of fiddling.

Ragged-right paragraphs require different H&Js than justified paragraphs. However, the typographer has the same goal — to keep the overall rhythm as even as possible, so that the author's ideas can flow smoothly into the reader's brain.

THE TYPOGRAPHER CAN ADJUST THE HYPHENATION RULES TO SUIT THEIR DOCUMENT.

Usually we don't Hyphen-ate capitalized words.

BUT IT CAN BE OK TO HYPHENATE FREQUENTLY OCCURRING PROPER NOUNS AFTER THEIR FIRST APPEARANCE IN THE TEXT.

It's bad to hyphenate short words.

ALWAYS LEAVE AT LEAST 3 LETTERS BEFORE A HYPHEN AND MAKE SURE TO LEAVE AT LEAST 3 LETTERS AFTER A HYPHEN.

Narrow columns make hyphenation hard to avoid. The typographer tries to prevent multiple hyphenated lines in a row.

WHEN SETTING ENGLISH, TRY TO AVOID MORE THAN 3 HYPHENATED LINES IN A ROW. TOO MANY HYPHENS CAN DISTRACT THE READER.

Hyphenation

Hyphenation, in the context of typography, is when you break a word at the end of a line in order to get the line to fit the way you want. The rules for proper hyphenation can vary from language to language, and the breaking is usually done between syllables.

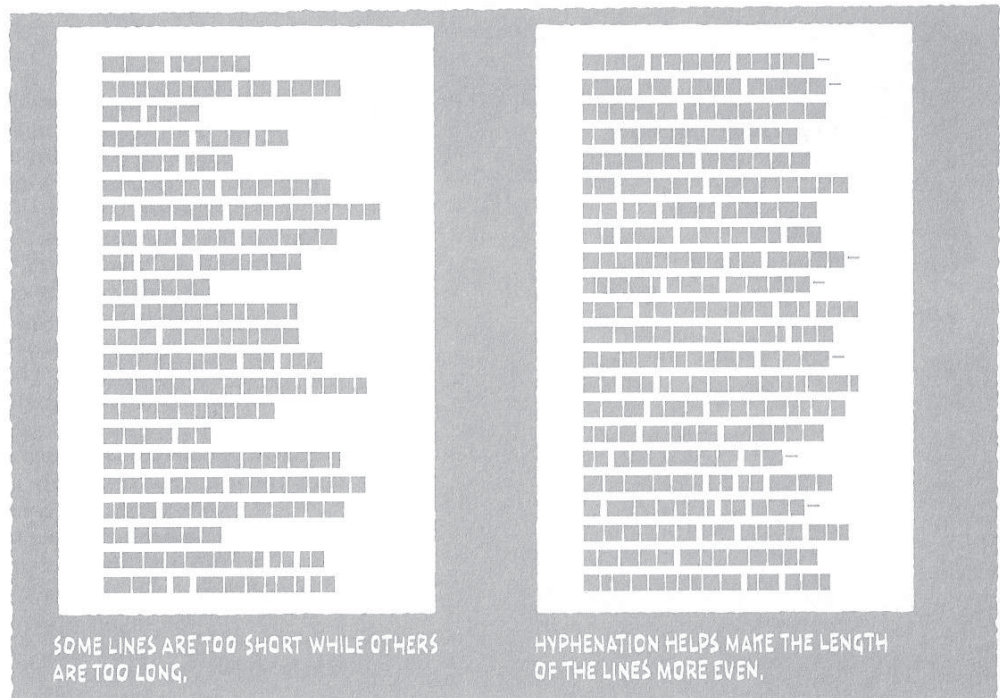
A sophisticated typesetting application will contain built-in hyphenation dictionaries that are used to determine where it can break words and where it can't. These dictionaries usually do a fairly good job. The typographer's job is then to fine-tune the settings, check the results, and make manual corrections when needed.

There are some hyphen-phobic typographers who think hyphenation is never acceptable and fixate on eliminating hyphens at all costs. A more practical approach is to think of hyphenation as a necessary evil. Well-tuned hyphenation settings help keep the text flowing smoothly. Spacing that's too irregular is usually more disruptive to the reader than a hyphenated word.

In narrow columns, the typographer needs to be more lenient with the hyphenation in order to maintain an even pace. This may mean allowing for a higher rate of hyphens in consecutive lines, and even letting proper nouns be hyphenated. This can feel disruptive to some readers, but it's a reasonable compromise.

HYPHENATION IS A COMPROMISE. THE TYPOGRAPHER USES IT TO MAINTAIN THE TEMPO AND KEEP THE RHYTHM OF THE TEXT AS EVEN AS POSSIBLE.

HYPHENATION CAN HELP
KEEP THE TEMPO MOVING AT
A REGULAR PACE. NARROW
COLUMNS CAN ESPECIALLY
BENEFIT FROM HYPHENATION.



SOME LINES ARE TOO SHORT WHILE OTHERS
ARE TOO LONG.

HYPHENATION HELPS MAKE THE LENGTH
OF THE LINES MORE EVEN.

Geschwin- digkeits- über- schreitung

HYPHENATION RULES ARE DIFFERENT FOR DIFFERENT LANGUAGES.
ABOVE IS AN EXAMPLE OF A LONG COMPOSITE WORD FROM GERMAN.
IT MEANS 'DRIVING OVER THE SPEED LIMIT'.

Ragged Text

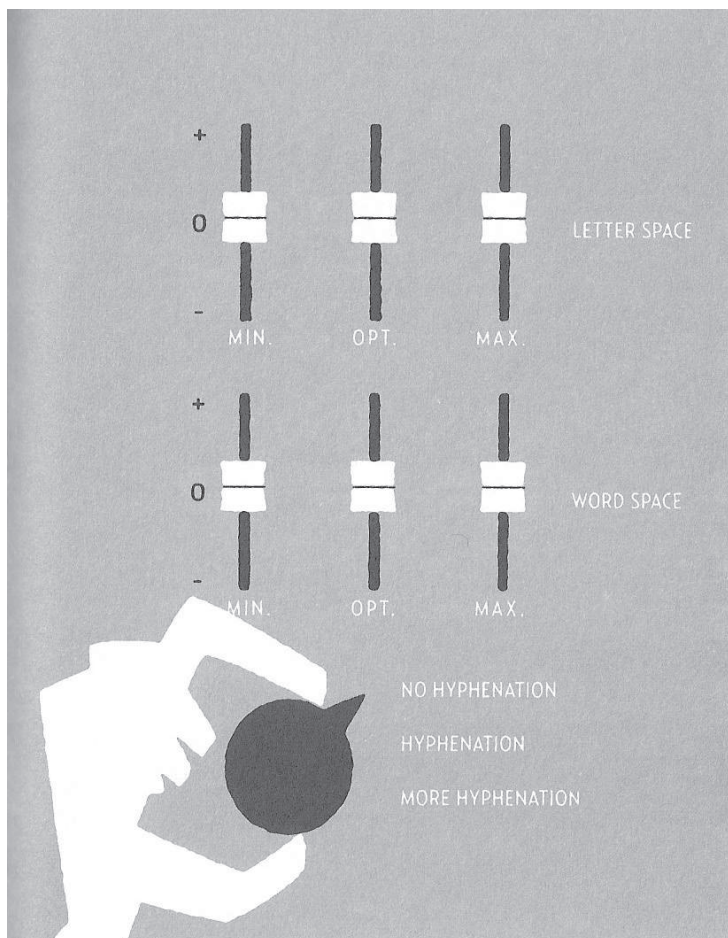
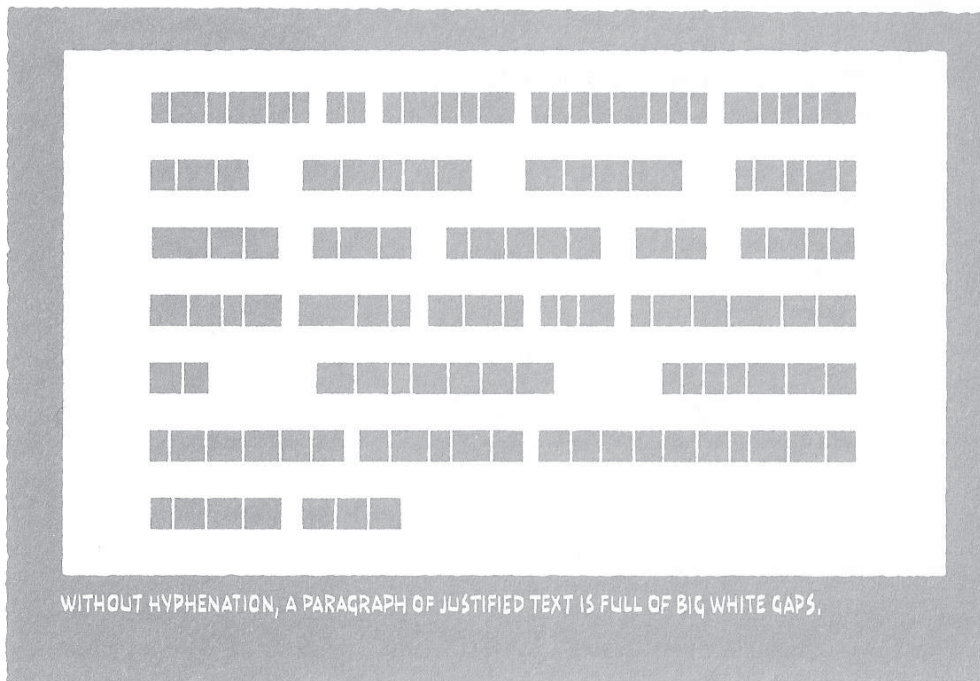
In ragged text, the typographer can decide whether or not to allow hyphenation. That decision is usually based on column width.

In a narrow column, hyphenation helps keep the line lengths fairly consistent. Without it, a long word can force a premature line break that will distract the reader. Hyphenation lets the rhythm of the line breaks remain more steady.

Hyphenation usually isn't necessary in a wide column of ragged text. The lines are longer and less likely to be upset by a long word or two. The rhythm of the line breaks will remain fairly steady even with the hyphenation turned off, so there's no need to distract the reader by breaking up words. Words that are already hyphenated, like *quick-frozen*, can still be broken.

It works differently in languages that have long composite words, such as German. In such cases, hyphenating becomes almost inevitable, even in wide ragged columns.

JUSTIFIED TEXT ALMOST ALWAYS
BENEFITS FROM HYPHENATION.



Justified Text

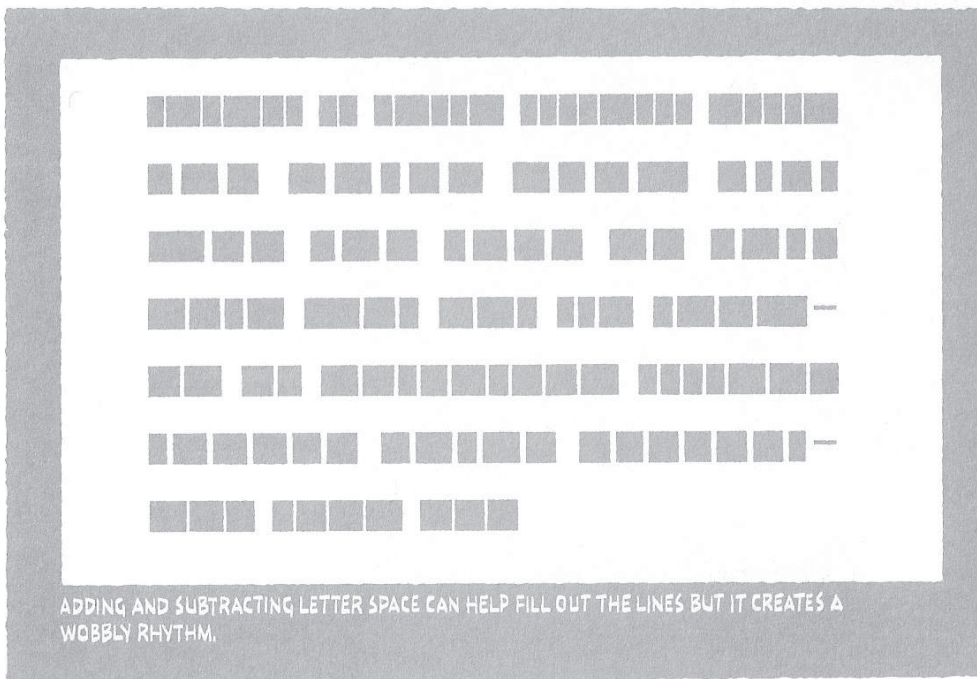
In justified paragraphs (like the ones in this book), H&Js are more complicated. In addition to making decisions about hyphenation, the typographer must also control the spacing within each line.

Without hyphenation, it would be impossible to avoid spacing and tempo problems in a paragraph of justified text. Well, not impossible — the typographer could rewrite the text to fill out the lines, but most authors get a little touchy about that. Even in wide columns, hyphenation is necessary when setting justified text.

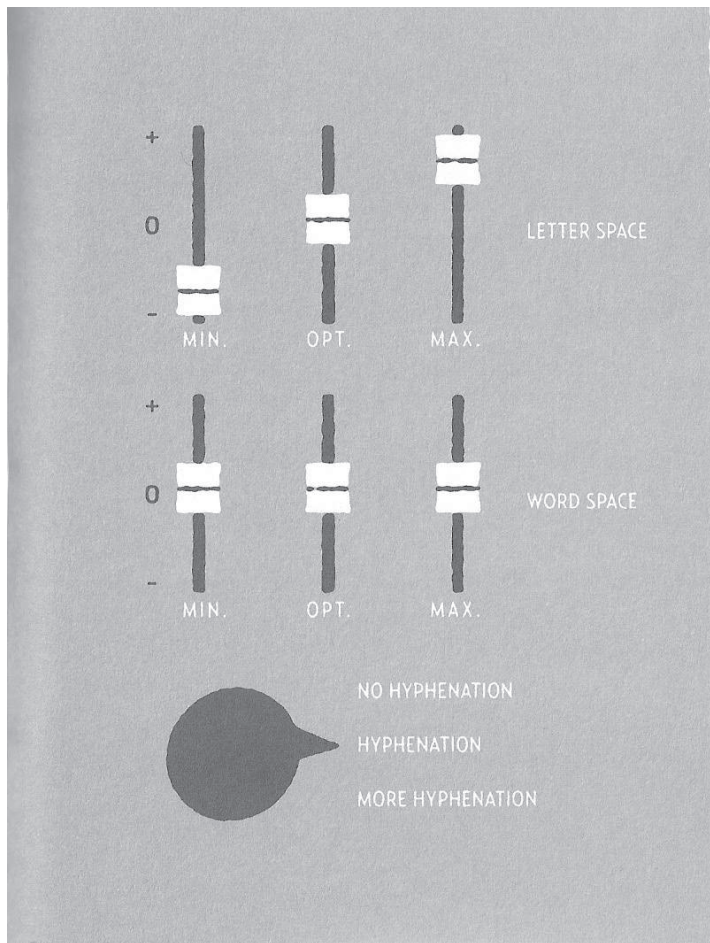
In ragged text, the spacing is consistent throughout a paragraph. In justified text, the spacing varies from line to line in order to force each line to be the same length as the others. This spacing variation can happen by adjusting the letter spacing, the word spacing, or a little of both. The typesetting system does the math and makes these spacing adjustments automatically. Some systems will even adjust the counter space by horizontally scaling the letters.

USING THE H&J SETTINGS, THE TYPOGRAPHER FINE-TUNES THE LETTER SPACE, WORD SPACE, AND HYPHENATION.

CHANGES TO THE LETTER SPACE CAN UNDO CAREFUL ADJUSTMENTS TO THE SPACING.



ADDING AND SUBTRACTING LETTER SPACE CAN HELP FILL OUT THE LINES BUT IT CREATES A WOBBLY RHYTHM.

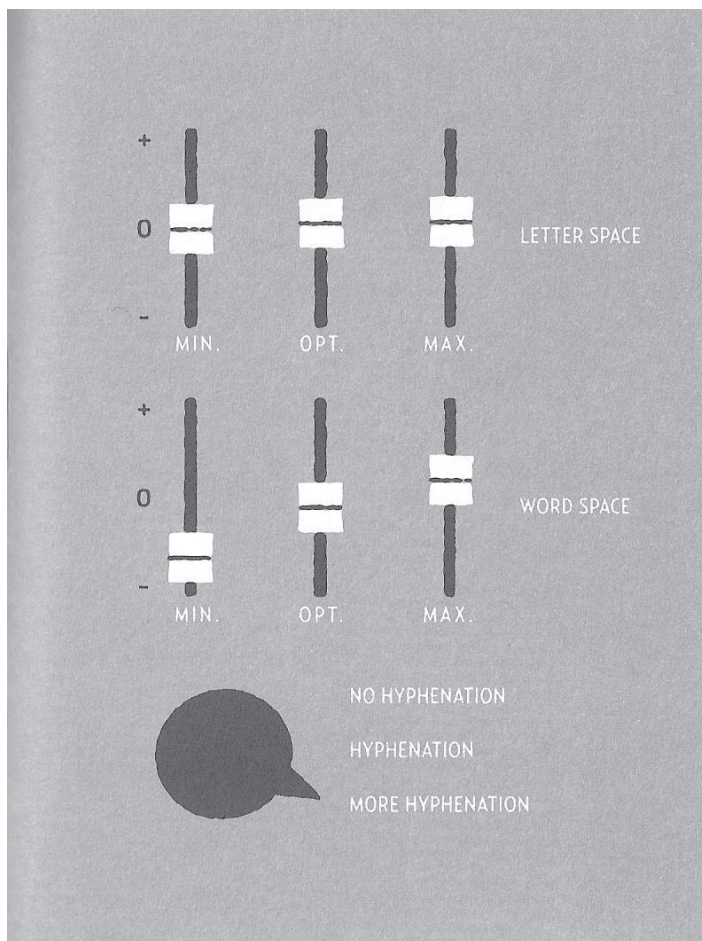
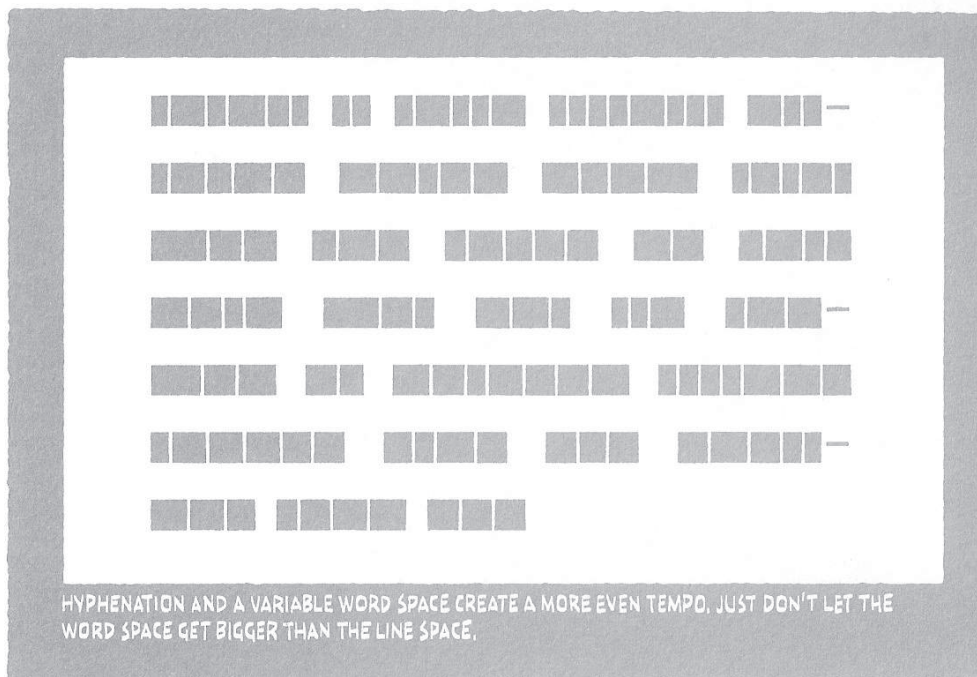


Justification Settings

The justification settings affect how a typesetting system does its math and makes spacing adjustments to a line of text. The typographer's job is to tune these settings by determining the minimum, optimum and maximum allowable sizes for the letter space and word space. The goal is to keep the overall rhythm of the paragraph as even as possible.

Letter space is a small interval and as such it is very important to the overall rhythm of the text. Tracking adjustments to a font's built-in spacing are always made with care. However, letter spacing can also be affected by the justification settings. This means that the justification settings have the potential to undo the careful work that went into determining the appropriate tracking on the overall paragraph settings. Therefore, in the justification settings, it's best to set the optimum letter space to 0%. This setting helps preserve the typographer's previous decision about letter space.

KEEPING THE TEMPO EVEN IN JUSTIFIED TYPE IS A BALANCING ACT BETWEEN HYPHENATION AND CHANGES TO THE SPACING.



Minimum, Optimum, and Maximum

In addition to the optimum letter space, the typographer can set the minimum and maximum letter space. These parts of the justification settings determine how much the letter space can grow or shrink as part of the system's math for justifying the paragraph. Experienced typographers know that subtle changes in letter space can have a big impact. It's a small space. If the letter space changes even a bit from line to the next, it will disrupt the flow of the text. Therefore, it is recommended to also keep the minimum and maximum allowable letter space set to 0%. This will keep the letter space consistent throughout the paragraph.

Word space is a larger interval, so it's less critical to the overall flow of the paragraph. After the correct optimum word space is determined, the minimum and maximum can be set to allow the word space to shrink or grow in order to force the lines to be the same length. The minimum word space can be allowed to get quite small. The greater danger is in letting the maximum word space get too big. Big gaps between words can upset the paragraph's spacing hierarchy and the reader will have a harder time following the lines of text across the page. Gaps in multiple consecutive lines can form distracting white rivers of space that trickle vertically through a paragraph.

less no one has yet forgotten. Not to mention rumours which agitated the maritime population and excited the public mind, even in the interior of continents, seafaring men were particularly excited. Merchants, common sailors, captains of vessels, skippers, both of Europe and America, naval officers of all countries, and the Governments of several States on the two continents, were deeply interested in the matter.

For some time past vessels had been met by "an enormous thing," a long object, spindle-shaped, occasionally phosphorescent, and infinitely larger and more rapid in its movements than a whale.

The facts relating to this apparition (entered in various log-books) agreed in most respects as to the shape of the object or creature in question, the untiring rapidity of its movements, its surprising power of locomotion, and the peculiar life with which it seemed endowed. If it was a whale, it surpassed in size all those hitherto classified in science. Taking into consideration the mean of observations made at divers times—rejecting the timid estimate of those who assigned to this object a length of two hundred feet, equally with the exaggerated opinions which set it down as a mile in width and three in length—we might fairly conclude that this mysterious being surpassed greatly all dimensions admitted by the learned ones of the day, if it existed at all. And that it did exist was an undeniable fact; and, with that tendency which disposes the human mind in favour of the marvellous, we can understand the excitement produced in the entire world by this supernatural apparition. As to classing it in the list of fables, the idea was out of the question.

On the 20th of July, 1866, the steamer Governor Higginson, of the Calcutta and Burnach Steam Navigation Company, had met this moving mass five miles off the east coast of Australia. Captain Baker thought at first that he was in the presence of an unknown sandbank; he even prepared to determine its exact position when two columns of water, projected by the mysterious object, shot with a hissing noise a hundred and fifty feet up into the air. Now, unless the sandbank had been submitted to the intermittent eruption of a geyser, the Governor Higginson had to do neither more nor less than with an aquatic mammal, unknown till then,

West India and Pacific Ocean Navigation Company. But this extraordinary creature could transport itself from one place to another with surprising velocity; as, in an interval of three days, the Governor Higginson and the Columbus had observed it at two different points of the chart, separated by a distance of more than seven hundred nautical leagues.

Fifteen days later, two thousand miles farther off, the Helvetia, of the Compagnie-Nationale, and the Shannon, of the Royal Mail Steamship Company, sailing to windward in that portion of the Atlantic lying between the United States and Europe, respectively signalled the monster to each other in 42° 15' N. lat. and 60° 35' W. long. In these simultaneous observations they thought themselves justified in estimating the minimum length of the mammal at more than three hundred and fifty feet, as the Shannon and Helvetia were of smaller dimensions than it, though they measured three hundred feet over all.

Now the largest whales, those which frequent those parts of the sea round the Aleutian, Kulammak, and Ungullich islands, have never exceeded the length of sixty yards, if they attain that.

In every place of great resort the monster was the fashion. They sang of it in the cafes, ridiculed it in the papers, and represented it on the stage. All kinds of stories were circulated regarding it. There appeared in the papers caricatures of every gigantic and imaginary creature, from the white whale, the terrible "Moby Dick" of sub-arctic regions, to the immense kraken, whose tentacles could entangle a ship of five hundred tons and hurry it into the abyss of the ocean. The legends of ancient times were even revived.

Then burst forth the unending argument between the believers and the unbelievers in the societies of the wise and the scientific journals. "The question of the monster" inflamed all minds. Editors of scientific journals, quarrelling with believers in the supernatural, spilled seas of ink during this memorable campaign, some even drawing blood; for from the sea-serpent they came to direct personalities.

During the first months of the year 1867 the question seemed buried, never to revive, when new facts were brought before the public. It was then no longer a scientific problem to be solved, but a real danger seriously to be avoided. The

finding herself during the night in 27° 30' lat. and 72° 15' long., struck on her starboard quarter a rock, marked in no chart for that part of the sea. Under the combined efforts of the wind and its four hundred horse power, it was going at the rate of thirteen knots. Had it not been for the superior strength of the hull of the Moravian, she would have been broken by the shock and gone down with the 237 passengers she was bringing home from Canada.

The accident happened about five o'clock in the morning, as the day was breaking. The officers of the quarter-deck hurried to the after-part of the vessel. They examined the sea with the most careful attention. They saw nothing but a strong eddy about three cables' length distant, as if the surface had been violently agitated. The bearings of the place were taken exactly, and the Moravian continued its route without apparent damage. Had it struck on a submerged rock, or on an enormous wreck? They could not tell; but, on examination of the ship's bottom when undergoing repairs, it was found that part of her keel was broken.

This fact, so grave in itself, might perhaps have been forgotten like many others if, three weeks after, it had not been re-enacted under similar circumstances. But, thanks to the nationality of the victim of the shock, thanks to the reputation of the company to which the vessel belonged, the circumstance became extensively circulated.

At seventeen minutes past four in the afternoon, whilst the passengers were assembled at lunch in the great saloon, a slight shock was felt on the hull of the Scotia, on her quarter, a little aft of the port-paddle.

The Scotia had not struck, but she had been struck, and seemingly by something rather sharp and penetrating than blunt. The shock had been so slight that no one had been alarmed, had it not been for the shouts of the carpenter's watch, who rushed on to the bridge, exclaiming, "We are sinking! we are sinking!" At first the passengers were much frightened, but Captain Anderson hastened to reassure them. The danger could not be imminent. The Scotia, divided into seven compartments by strong partitions, could brave with impunity any leak. Captain Anderson went down immediately into the hold. He found that the sea was pouring into the fifth compartment; and the rapidity of the influx showed that the force of the water was

H&J Strategies

Different typographers may have different strategies for setting the H&Js. Some consider allowing a slight amount of variation in letter space to be acceptable, while others (including myself) think it's best avoided. Conversely, some hyphenation is usually all right with me, but others think it's sloppy. These varying opinions are shaped by the different kinds of documents and texts each typographer works with. A newspaper designer will naturally have a different point of view than a book designer. As you gain experience setting type, you'll probably develop your own approach.

If the paragraph settings and H&Js are done well, automated typesetting can be fairly successful in setting a paragraph with a good, even flow. However, the typographer still needs to check each paragraph. Narrow columns especially benefit from someone with an experienced eye taking the time to make a manual line break or spacing adjustment where needed.

Finally, the typographer always needs to remain flexible. While it's important to have strong opinions about how type should be set, sometimes compromises must be made. It's the only way to get the story told. In fact, readers with sharp eyes may find examples in this book where my own recommendations couldn't be followed to the letter.

CONGRATULATIONS, you've reached the end of the book. Or, if you're like me and start a book by reading the conclusion first — *welcome!*

The material in this book covers a lot of ground, from how type works and how we read, to tips on fine-tuning a paragraph of text. It isn't critical that the beginning typographer master all of it immediately. The best way to learn is by setting as much type as you can. While you do that, think about the things you've read in this book, and consider re-reading them.

A beginner can find all the different kinds of white space within a paragraph overwhelming and even feel paralyzed by the knowledge that adjusting one kind of white space will have an effect on all the others. It can seem easier to simply focus on the black letters and not worry about the white space at all.

But you're not alone: typographers can think of composing type as a collaboration between themselves and their chosen typeface. The typeface's built-in spacing forms the basic structure of the paragraph and is a fine starting point. Then you can tune the paragraph settings to suit the text and the document.

CONCLUSION

Typographic adjustments are often subtle, especially to the untrained eye. But good typography does more than make text legible. It can add depth and character to the presentation. In this sense, you can think of the typography as another part of the story.

Being able to see and understand the structure of a paragraph of printed text is a good foundation for learning the art of typography. The typographer's understanding of white space extends to the rest of the page, from the gutter between columns to the margins that surround them. Then, when setting different kinds of documents and working in different media, the typographer acquires additional skills and vocabulary. All these things build on this foundation.

The world is full of typography. If you can teach yourself how to look, type is everywhere and ready to be enjoyed.

COLOPHON

The body text in this book is set in Ibis Text. The headings are set in Scout. Both Ibis and Scout are original typeface series drawn by Cyrus Highsmith, available from Font Bureau. The captions were hand-lettered.

The book was designed by Cyrus Highsmith with help from Kent Lew. They used Adobe InDesign. Cyrus Highsmith drew the illustrations using all sorts of tools including a pencil, Adobe Photoshop, and RoboFont.